Unit 2 Numerical Control

A Definition

Numerical Control is a system in which actions are controlled by the direct insertion of **numerical data** at some point. The system must automatically interpret at least some portion of the data

HISTORICAL DEVELOPMENT

- 15th century - machining metal.
- 18th century - industrialization, production-type machine tools.
- 20th century - F.W. Taylor - tool metal - HSS

  Automated production equipment -

  Screw machines

  Transfer lines

  Assembly lines

  using cams and preset stops

  Programmable automation -

  NC

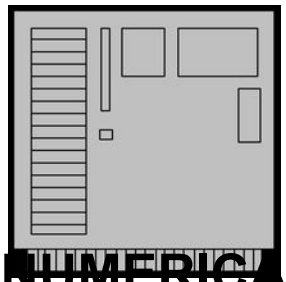
  PLC

  Robots



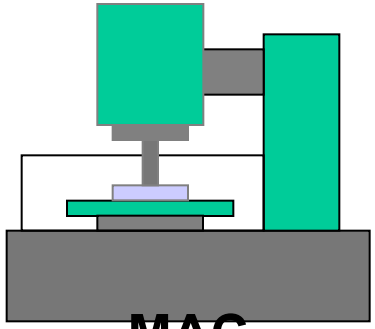
MANUFACTURING OPERATOR

NUMERICAL DATA (NC CODE)



NUMERICAL CONTROLLER

Drive Control



MACHINE UNIT

PROCESSED PART



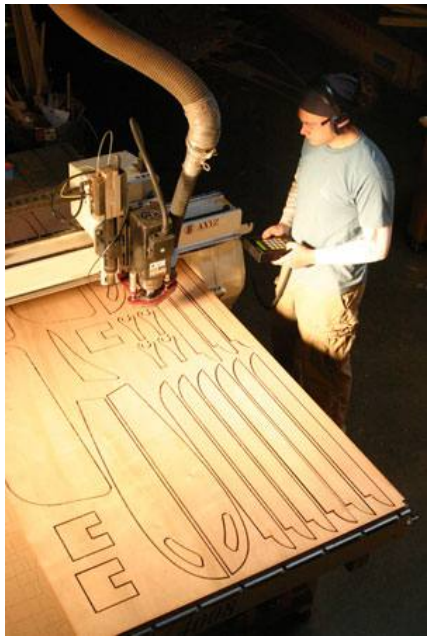


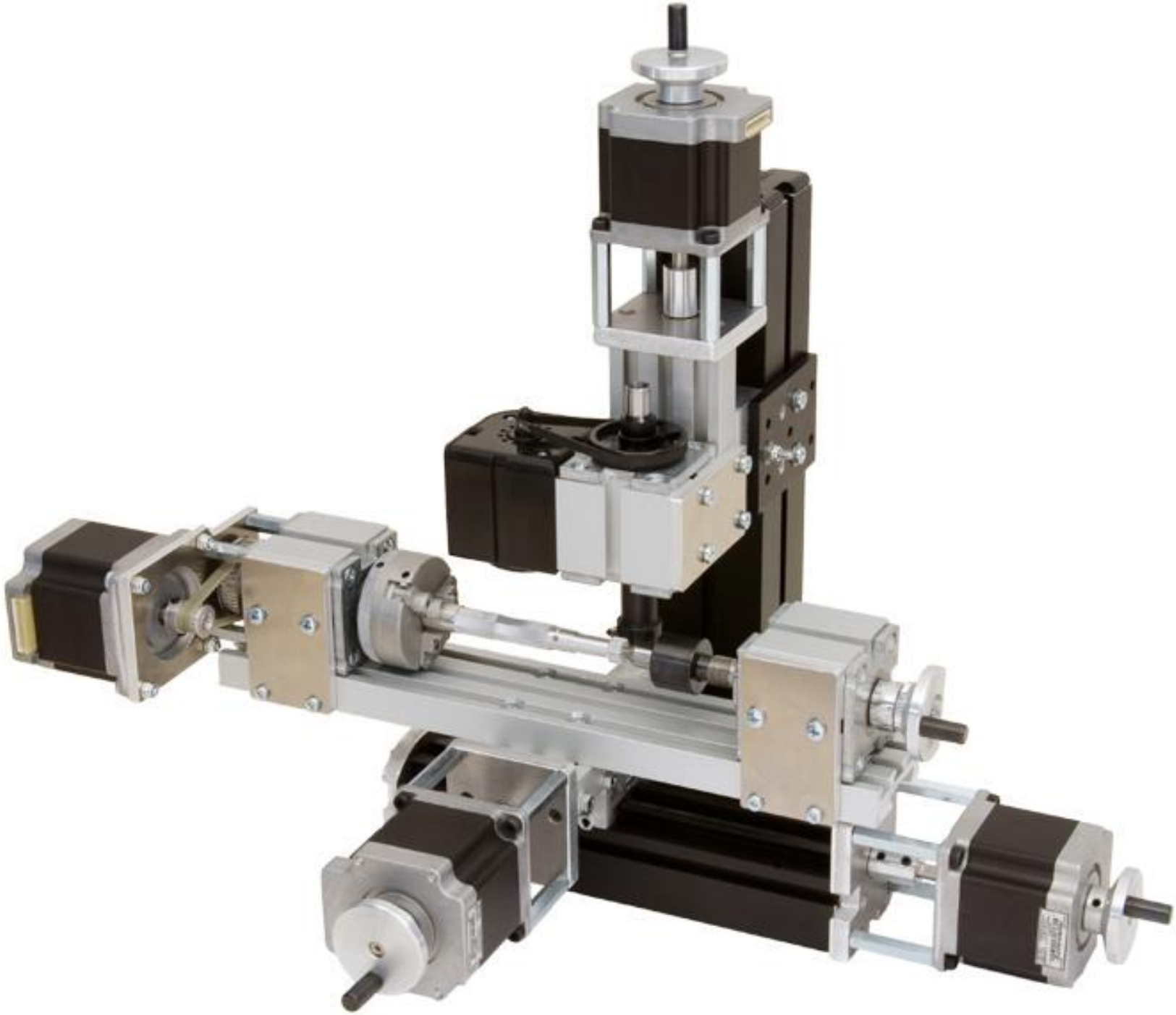






CNC Laser Cutting Machine, Capacity upto 28 mm Thickness.





NC Procedure

1. Process Planning: the engineering drawing of the workpart must be in terms of the manufacturing process to be used. Planning is a preparatory step before manufacturing, which determines the sequence of operations or processes needed to produce a part or an assembly. This step is more important in job shops, where one-of-a-kind products are made or the same product is made infrequently.
2. Part programming:which describe the work, which has to be done on a **part**, in the form required by a computer under the control of computer numerical control (CNC) software. It is the task of preparing a program sheet from a drawing sheet

3.Tape preparations:Holes are punched in the tape using a tape punch machine. Tape preparation is accomplished either manually or using the computer. In the manual method, the tape punch machine operates in conjunction with a typewriter-like machine

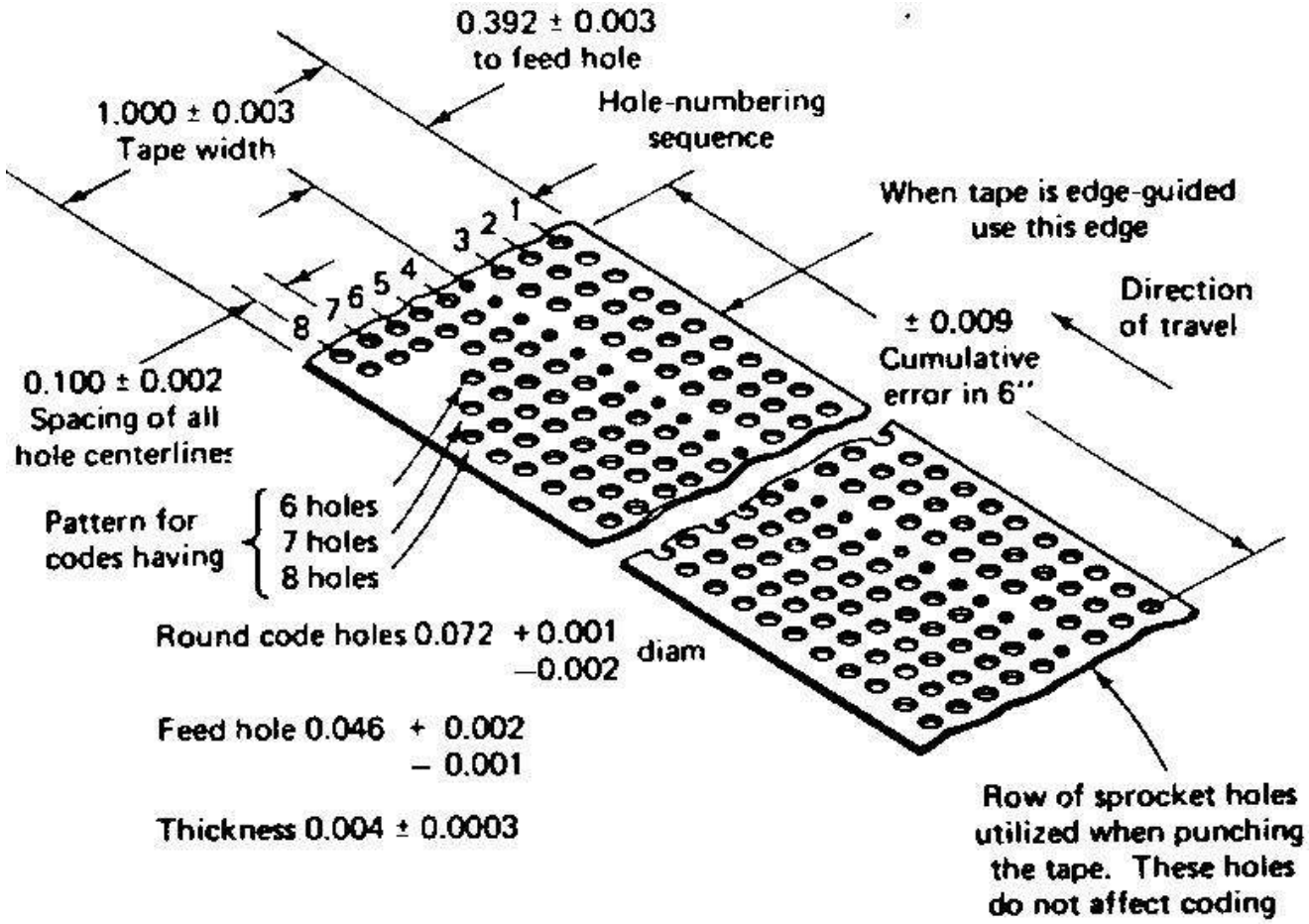


FIGURE 9.1 Numerical control punched tape format as standardized by the Electronics Industries Association (EIA). (Reprinted from Childs [1].)

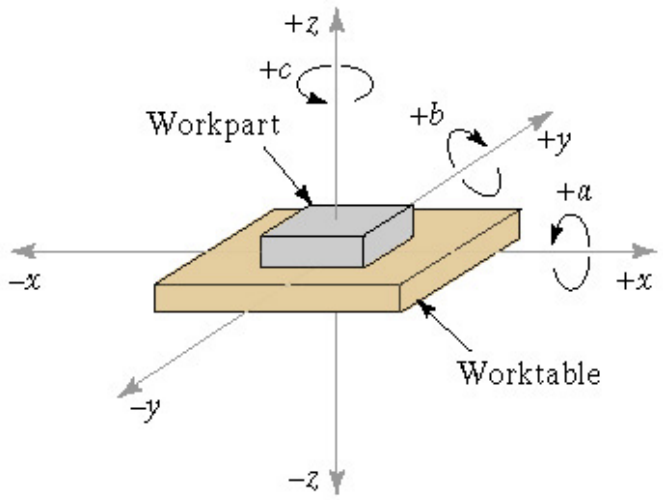
4.Tape verification: After the punched tape is prepared, a method is usually provided for checking the accuracy of the tape. Sometimes the tape is checked by running it through a computer program which plots the various tool movements.

5. Production: the final step in the NC producer is to use the NC Tape in production. This includes ordering the raw materials, specifying and preparing the tooling and any special fixturing that may be required and setting up the NC machine tool for job.

NC Coordinate Systems

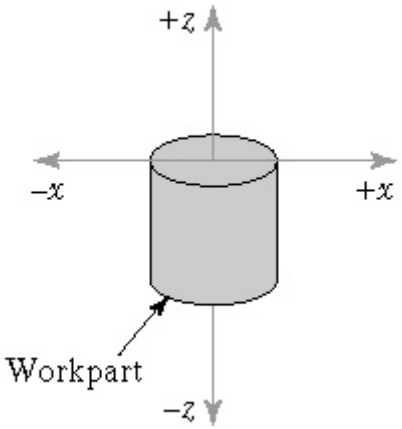
For flat and prismatic (block-like) parts:

- Milling and drilling operations
- Conventional Cartesian coordinate system
- Rotational axes about each linear axis



For rotational parts:

- Turning operations
- Only *x*- and *z*-axes



Absolute versus incremental positioning

Absolute positioning means that the tool locations are always defined in relation to the zero point. If a hole is to be drilled at a spot that is 8 in. above the x-axis and 6 in. to the right of the y-axis, the coordinate location of the hole would be specified as $x = +6.000$ and $y = +8.000$.

Incremental positioning means that the next tool location must be defined, with reference to the previous tool location. If in the previous drilling example, the previous hole had been drilled at an absolute position of $x = +4.000$ and $y = +5.000$, the incremental position instructions would be specified as $x = +2.000$ and $y = +3.000$ in order to move the drill to the desired spot. Figure 8.4 illustrates the difference between absolute and incremental positioning

Absolute versus incremental positioning

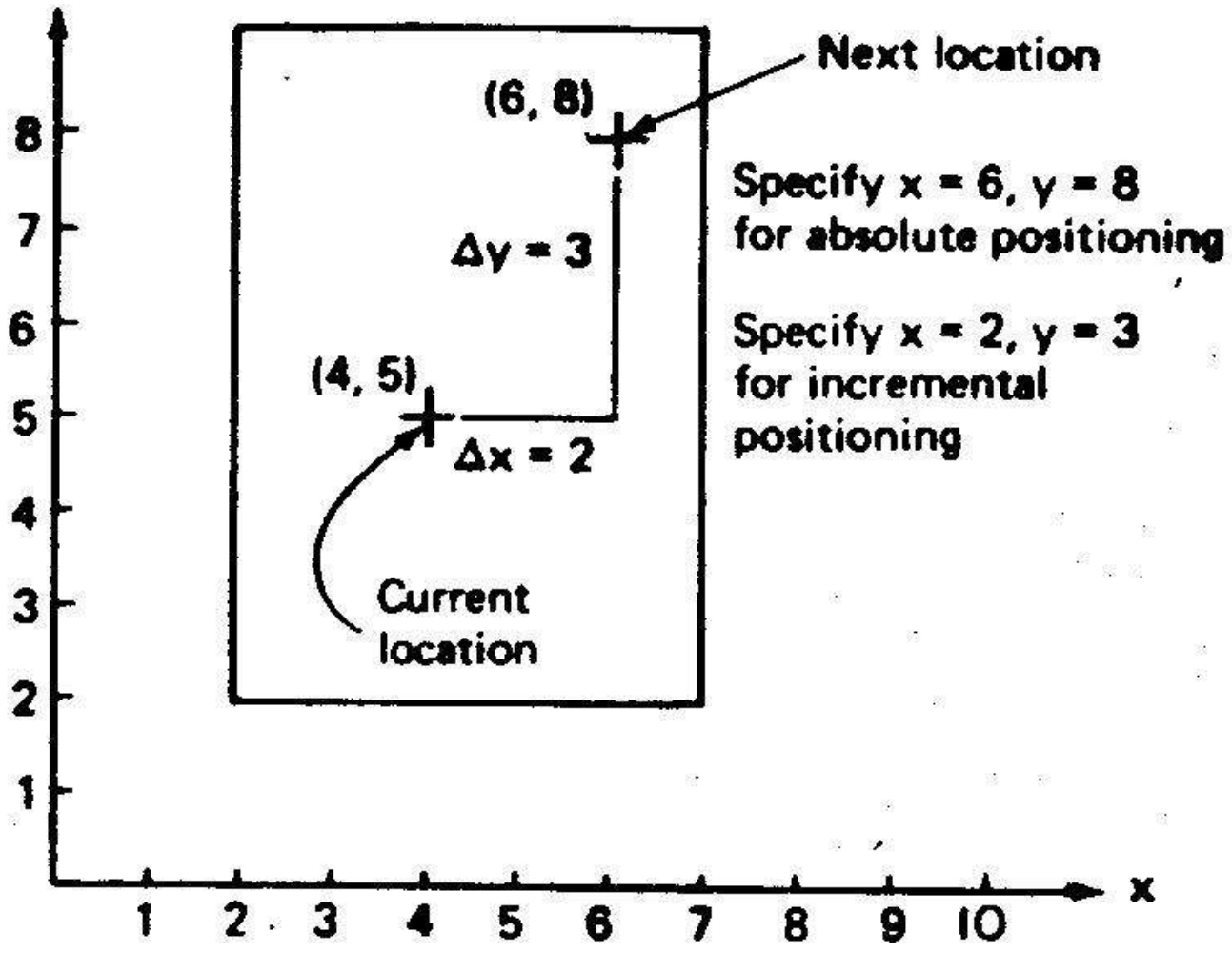
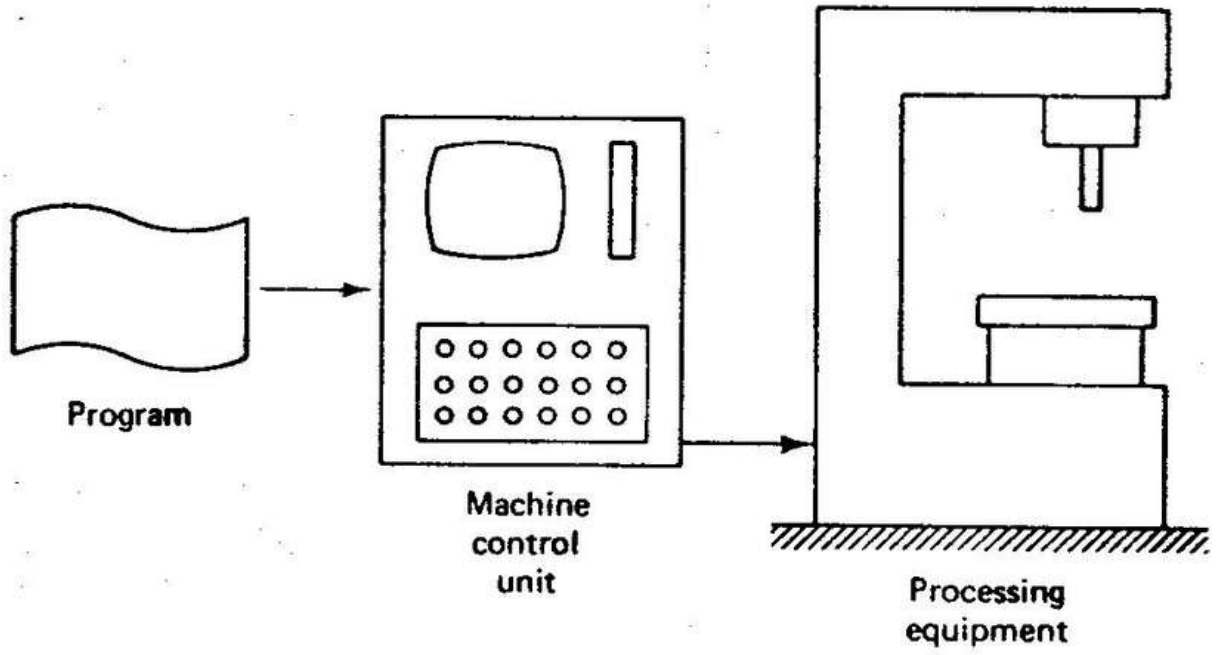


FIGURE 8.4 Absolute versus incremental positioning.

Elements of NC system

The basic components are:

1. Program of instructions
2. Machine control unit
3. Machine tool or Processing equipment or other controlled processes



1. Program of Instructions

- The program of instructions is a complete step-by-step set of instructions that tells the machine tool what to do. It is coded on some type of input medium in a numeric or symbolic form that can be interpreted by the controller unit.
- There are two other methods of input in NC systems that should be mentioned. The first is manual entry of instructional data for the controller unit.
- This method is called manual data input, decreased MDI, and is only suitable for relatively simple jobs where the command will not be repeated. Other methods of input are through direct links with computers. This is termed as direct numerical control or DNC.
- A program of instructions is prepared by a part programmer. The programmer's job is to provide a set of detailed instructions by which to order the processing steps. For the machining operation, the processing step includes relative movement between the cutting tool and the workpiece.

2. Machine Controller Unit

- It includes electronics and hardware that read and interpret the program of instructions and convert it into mechanical functions of machine tools. This control unit controls the passage of equipment, feeds, tool-changes and other functions.
- Common components of a traditional NC controller unit include tape readers, a data buffer, signal channels for machine tools, feedback channels from machine tools. Series controls to coordinate the overall operation of the foregoing elements.
    1. Tape reader
    2. Signal output unit
    3. Sequence coordinator

Classification of NC Systems

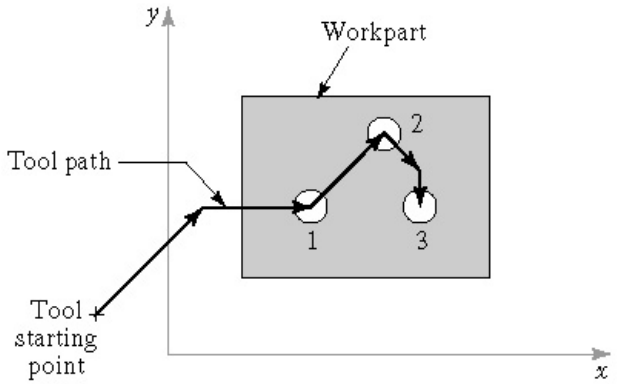
The classification of NC machine tool systems can be done in three ways:

1. According to the type of machine: Point-to-point, Straight cut and contouring (continuous path)
2. According to the programming method: incremental versus absolute
3. According to the type of control loops: Open-loop versus closed-loop

Motion Control Systems

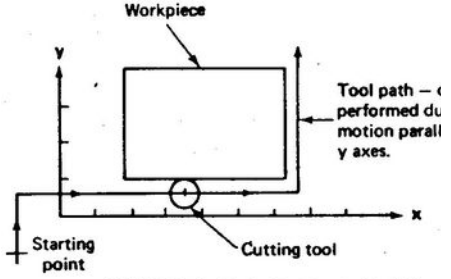
Point-to-Point systems

- Also called position systems
- System moves to a location and performs an operation at that location (e.g., drilling)
- Also applicable in robotics



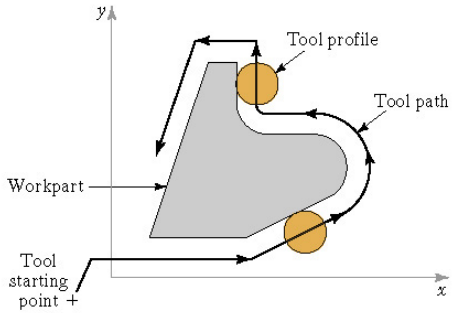
Straight Cut

- The NC systems, in which the tool works along a straight line in the direction of a major coordinate axis, such as along the direction of feed during turning, boring or milling operation at a controlled rate, are known as Straight line control system.



Continuous path systems

- Also called contouring systems in machining
- System performs an operation during movement (e.g., milling and turning)



Absolute versus incremental positioning

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Absolute versus incremental positioning

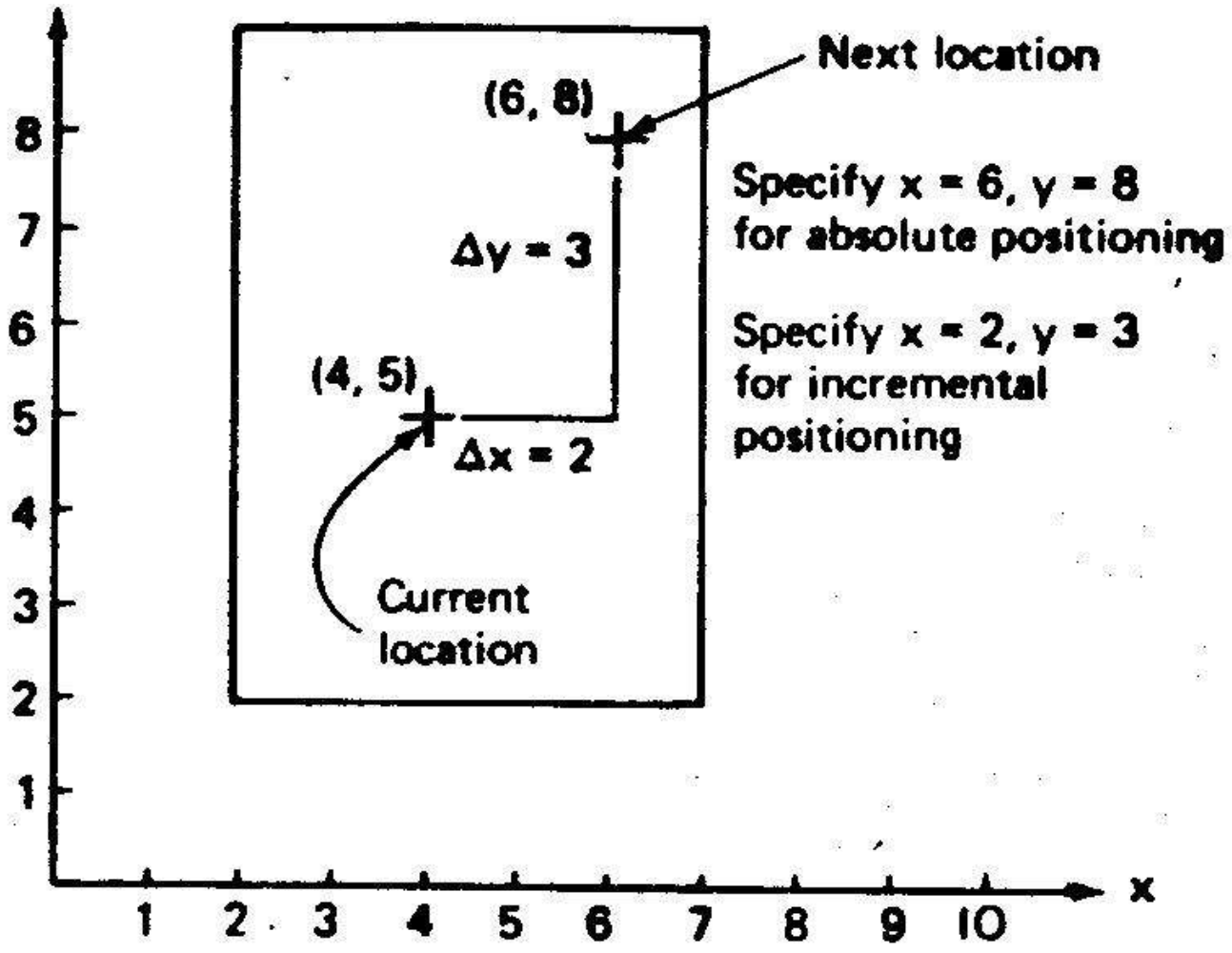


FIGURE 8.4 Absolute versus incremental positioning.

Classified by the servo system

- Open-loop control system means that there is no detection and feedback device. After power amplification by the driving circuit, the instruction signal from CNC device drives the machine table to move through the stepping motor, and the signal transmission is unidirectional.

- Closed-loop control system adopts a linear displacement detection device, which is installed on the moving parts or workbench of the machine tool. The actual displacement detected is fed back to the comparator of the CNC device, and is compared with the program instruction value. Control with the difference value until the difference value is zero.

Functions of the CNC

CNC is designed to perform many functions which either difficult or impossible to perform by a NC machine.

1. Machine Tool Control
2. In-process compensation
3. Improved programming and operating features
4. Diagnostic

1. Machine Tool Control

Controlling the machine tool is the primary function of a CNC. It includes the conversion of part program instruction into machine tool motions. This conversion is done through a computer interface and servo system.

2. In-process compensation

- It involves the real-time dynamic correction of errors in the machine tools motion during processing. Some features of in-process compensation are listed below
  Debugging the problem and adjusting for the error sensed by in-process probes and gauges.
- Recalculation of axis positions when inspection probe is locating the reference on work part.
- Adaptive control adjustment to the tool speed and feed.
- Compute the predicted tool life and select alternative tooling when indicated.
- Offset adjustment for tool radius and length.

3. Improved programming and operating features

- The flexibility of soft-wired control of CNC allows the introduction of many convenient programming and operation features. The main features are listed below Manual data input and editing of part program at machine - it permits the correction and optimization of part program.
- Graphic display tool path help to verify the tape.
- Support more than one system of units.
- Use of specially written subroutine program.
- Types of interpolation technique: Cubic, circular and parabolic interpolation.
- Large local storage system that having more than one program.

4.Diagnostic

CNC machines are very expensive and complicated. The complexity of machine required well trained and proficient personal for maintenance. The CNC machines are often equipped with the diagnostic capability to assist during the maintenance of system.

- One of the main features is that it identifies the reason for a problem and guide the maintenance personnel so that he/ she can repair the system very quickly.
- Second feature that it alerts the imminent failure of certain component, so the maintenance personnel can check/ replace the faulty component.
- Third possible function is that when a component fails, the diagnostic subsystem automatically disconnects the faulty component and activate the redundant component

Features of CNC

1. Program storage
2. Online part program and editing
3. Enhanced cutter compensation
4. Flexibility to modify control function
5. Easier maintenance of CNC via diagnostic software
6. Tool path generation
7. Program proving facilities
8. Complex interpolation
9. Background programming
10. Special canned cycles
11. Programmable logic controller features

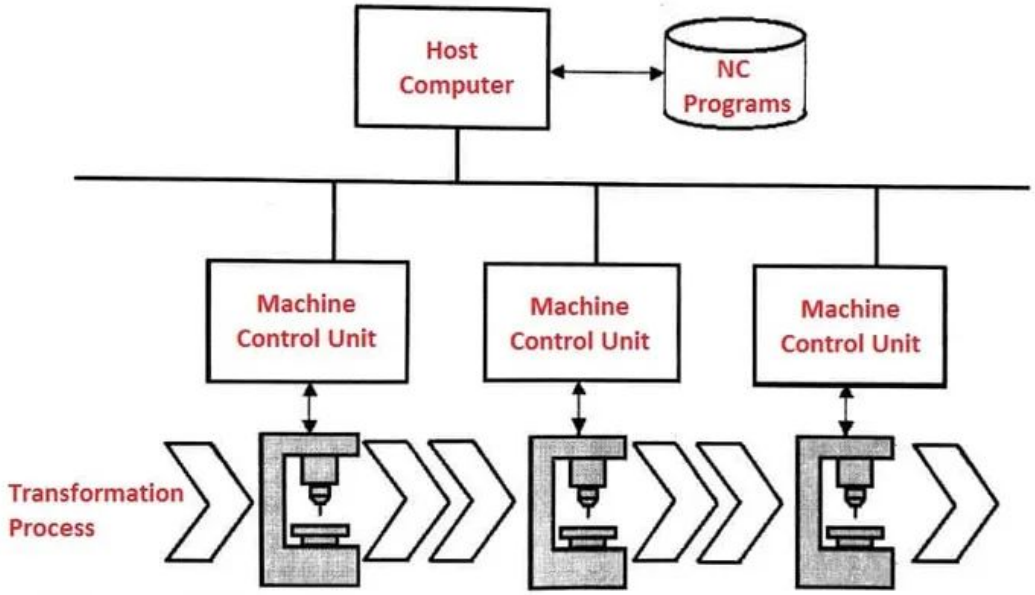
Industrial applications of CNC

CNC systems can be applied to a broad range of industrial applications. Following are some of the industries where CNC machining is used:

- Metal fabrication
- Electrical discharge machining
- Automotive
- Manufacturing
- Electronics
- Agriculture

Direct numerical control (DNC)

Direct numerical control (DNC) can be described as a various type of a manufacturing system in which that multiple NC Machine or CNC machines are remotely controlled from a host computer or DNC control of multiple machines tools by a single computer through a direct connection. It is shown in the diagram below.



DNC Machine System

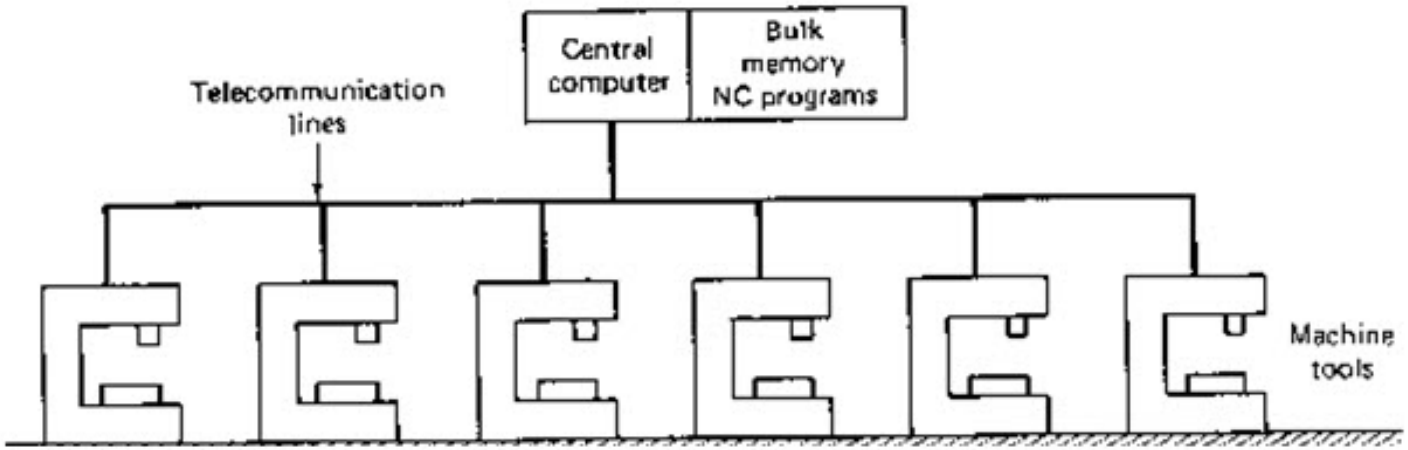
Functions of DNC System

Following are the functions of the DNC machine:

1. For use the central computer for storage and editing programs for all machine tools connected to it.
2. To give the stored programs to the connected machines on demand.
3. For to post-process part programs written in higher-level languages like APT.
4. To combine CAD with CAM by having a common database.
5. To provide a link between a central computer and various plant computers connected through modems and networks even though the plants may be placed several kilometres away from each other.

Components Used in DNC Machine

1. Central computer
2. Bulk memory for storing programs
3. Communication network
4. NC machine

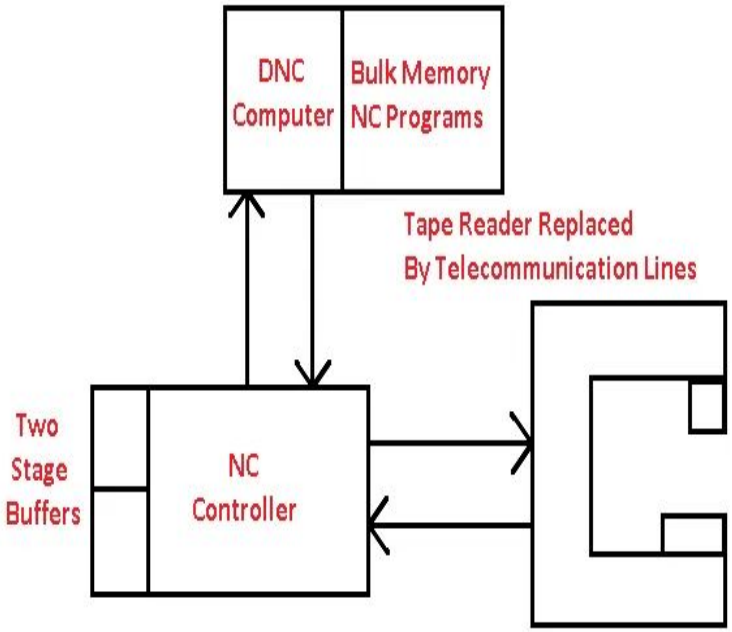


Types of DNC system

1. Behind the Tape Reader (BTR) system
2. Specialized MCU

1. Behind The Tape Reader (BTR) System

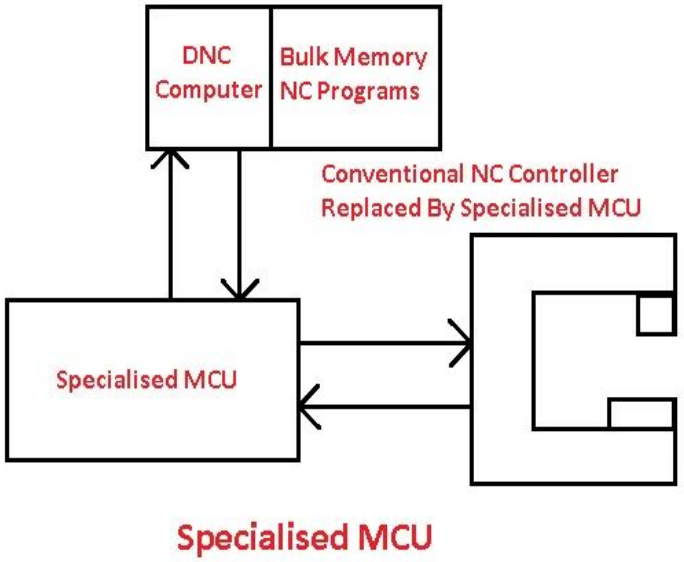
- In this type of system, the computer is connected directly to the regular NC controller unit. The operation of the system is similar to conventional NC, except for the source of command instructions.
- The controller unit employs two temporary storage buffers to get the blocks of instructions from the DNC computer and turn them into machine operations.
- The one buffer is getting a block of data, the other is providing control instructions to the specific machine tool. This system cost is very low.



Behind The Tape Reader (BTR) System

2. Specialised MCU

- In specialised MCU system, replace the normal controller unit with the special machine control unit.
- The special control unit is created to help communication between machine tools and computers.
- The specialized MCU configuration achieves a better balance between the accuracy of interpolation and the faster removal rate of the metal than is usually possible with the BTR system.



Specialised MCU

Advantages of DNC System

1. The DNC rejects the use of tape readers, which are absolutely the weakest component of the NC system.
2. Time-sharing by central control makes it possible to keep close control over the entire machine shop.
3. The huge memory of DNC allows it to store a large number of part programs for subsequent use. It also receives the memories of NC control unit.
4. Presence of a central bulk memory allows the same program to be run on different machines at the same time without duplicating it at individual places.

Disadvantages of DNC System

1. DNC uses a central control and in an event of computer failure, the complete activities of the machine shop would come to a standstill.
2. DNC is expensive and its use is practical in areas where high automation is required.

PART PROGRAMMING

Methods of NC Part Programming The tape can be prepared for submission to the MCU using any of several different methods of NC part programming. NC programming represents one of the elements in the broader procedure called process planning. We defined process planning as a function of manufacturing engineering in which the sequence of individual production operations for making the part are planned. We assume here that a portion of the processing is to be done on one or more NC machines. For those machines, the program must be prepared.

The part programming methods include a variety of procedures ranging from highly manual to highly automated:

1. Manual part programming
2. Computer assisted part programming
3. Manual data input
4. NC programming using CAD/CAM
5. Computer automated part programming

Manual part programming

In manual part programming, the processing instructions are documented on a form called a part program manuscript. The manuscript is a listing of the positions of the tool relative to the work piece that the machine must follow in order to perform the processing. The listing may also include other commands such as speeds, feeds, tooling, and so on. A punched tape is then prepared directly from the manuscript.

To be able to create a part program manually, we, need the following information :

(a)Knowledge about various manufacturing processes and machines.

(b)Sequence of operations to be performed for a given component.

(c)Knowledge of the selection of cutting parameters.

(d) Editing the part program according to the design changes.

(e) Knowledge about the codes and functions used in part programs.

Computer assisted part programming

In computer-assisted part programming, much of the tedious computational work required in manual programming is performed by the computer.

For complex work part geometries or jobs with many processing steps, use of the computer results in significant savings in the part programmer's time. When computer assisted part programming is used, the programmer prepares the set of processing instructions in a high-level computer language.

For complex jobs, this computer language is much easier to use than the lower-level coding required in manual part programming. The high-level language commands are interpreted by the computer, and the required calculations and data processing are accomplished to prepare the NC program for the tape reader (or other input device).

Manual data input

Manual data input (MDI) is a procedure in which the NC program is entered directly into the MCU at the site of the processing machine. Consequently, the use of the punched tape is avoided, and the programming procedure is simplified to permit machine operators rather than part programmers to do the programming.

NC programming using CAD/CAM

NC part programming using CAD/CAM is an advanced form of computer-assisted part programming in which an interactive graphics system equipped with NC programming software is used to facilitate the part programming task. The term CAD/CAM means computer-aided design and computer-aided manufacturing. In this method the programmer works on a CAD/CAM workstation to enter the machining commands. The actions indicated by the commands are displayed on the graphics monitor, which provides visual feedback to the programmer. Also, certain portions of the programming cycle are automated by the NC programming software to reduce the total programming time required.

Computer automated part programming

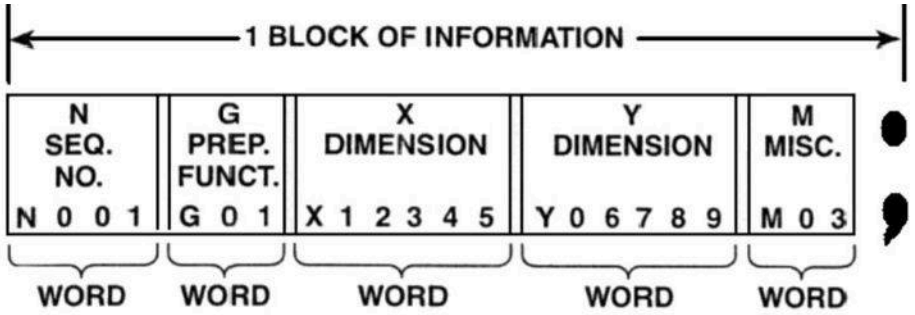
Computer-automated part programming extends the notion of automating certain portions of the. NC part programming procedure to its logical conclusion. It automates the complete part programming task using software that is capable of making logical and even quasi intelligent decisions about how the part should be machined

FUNDAMENTAL ELEMENTS FOR DEVELOPING MANUAL PART PROGRAMME

1. Type of Dimensioning System: We determine what type of dimensioning system the machine uses, whether an absolute or incremental dimensional system
2. Axis Designation The programmer also determines how many axes are availed on machine tool. Whether machine tool has a continuous path and point-to-point control system
3. NC Words The NC word is a unit of information, such as a dimension or feed rate and so on. A block is a collection of complete group of NC words representing a single NC instruction. An end of block symbol is used to separate the blocks. NC word is where all the machining data are compiled and where the data are translated in to a language, which can be understood, by the control system of the machine tool.

NC WORDS

Block of Information NC information is generally programmed in blocks of words. Each word conforms to the EIA standards and they are written on a horizontal line. If five complete words are not included in each block, the machine control unit (MCU) will not recognize the information; therefore the control unit will not be activated.



The words are as follows :

N001 – represents the sequence number of the operation.

G01 – represents linear interpolation.

X12345 – will move the table in a positive direction along the X-axis.

Y06789 – will move the table along the Y-axis.

M03 – Spindle on CW and

; – End of block.

Standard G and M Codes

The most common codes used when programming NC machines tools are G-codes (preparatory functions), and M codes (miscellaneous functions). Other codes such as F, S, D, and T are used for machine functions such as feed, speed, cutter diameter offset, tool number, etc.

G-codes are sometimes called cycle codes because they refer to some action occurring on the X, Y, and/or Z-axis of a machine tool. The G-codes are grouped into categories such as Group 01, containing codes G00, G01, G02, G03, which cause some movement of the machine table or head. Group 03 includes either absolute or incremental programming.

G-Codes (Preparatory Functions)

G00 Rapid positioning
G01 Linear interpolation
G02 Circular interpolation clockwise(CW)
G03 Circular interpolation counterclockwise (CCW)
G20 Inch input (in.)
G21 Metric input (mm)
G24 Radius programming
G28 Return to reference point
G29 Return from reference point
G32 Thread cutting

G40 Cutter compensation cancel
G41 Cutter compensation left
G42 Cutter compensation right
G43 Tool length compensation positive (+) direction
G44 Tool length compensation minus (-) direction
G49 Tool length compensation cancels
G 53 Zero offset or M/c reference
G54 Settable zero offset
G84 canned turn cycle
G90 Absolute programming
G91 Incremental programming

M-Codes (Miscellaneous Functions)

M-Codes (Miscellaneous Functions) M or miscellaneous codes are used to either turn ON or OFF different functions, which control certain machine tool operations. M-codes are not grouped into categories, although several codes may control the same type of operations such as M03, M04, and M05, which control the machine tool spindle. Some of important codes are given as under with their function

M-Codes (Miscellaneous Functions)

M00 Program stop
M02 End of program
M03 Spindle start (forward CW)
M04 Spindle start (reverse CCW)
M05 Spindle stop
M06 Tool change
M08 Coolant on
M09 Coolant off

M10 Chuck - clamping
M11 Chuck - unclamping
M12 Tailstock spindle out
M13 Tailstock spindle in
M17 Tool post rotation normal
M18 Tool post rotation reverse
M30 End of tape and rewind or main program end
M98 Transfer to subprogram
M99 End of subprogram

Tape Programming Format

Both EIA and ISO use three types of formats for compiling of NC data into suitable blocks of information with slight difference.

1. Word Address Format:

This type of tape format uses alphabets called address, identifying the function of numerical data followed. This format is used by most of the NC machines, also called variable block format. A typical instruction block will be as below :

N20 G00 X1.200 Y.100 F325 S1000 T03 M09

or

N20 G00 X1.200 Y.100 F325 S1000 T03 M09;

The MCU uses this alphabet for addressing a memory location in it.

Tape Programming Format

2. Tab Sequential Format

Here the alphabets are replaced by a Tab code, which is inserted between two words. The MCU reads the first Tab and stores the data in the first location then the second word is recognized by reading the record Tab. A typical Tab sequential instruction block will be as below

: >20 >00 >1.200 >.100 >325 >1000 >03 >09

3. Fixed Block Format

In fixed block format no letter address of Tab code are used and none of words can be omitted. The main advantage of this format is that the whole instruction block can be read at the same instant, instead of reading character by character. This format can only be used for positioning work only. A typical fixed block instruction block will be as below:

20 00 1.200 .100 325 1000 03 09

Coordinate Word

A co ordinate word specifies the target point of the tool movement or the distance to be moved. The word is composed of the address of the axis to be moved and the value and direction of the movement.
Example:

X150 Y-250
represents the movement to (150, -250). Whether the dimensions are absolute or incremental will have to be defined previously using G-Codes.

Parameter for Circular Interpolation

These parameters specify the distance measured from the start point of the arc to the center. Numerals following I, J and K are the X, Y and Z components of the distance respectively.

Spindle Function

The spindle speed is commanded under an S address and is always in revolution per minute. It can be calculated by the following formula in rpm:

$$\text{Spindle Speed} = \frac{\text{Surface cutting speed in m/min} \times 1000}{\pi \times \text{Cutter Diameter in mm}}$$

Feed Function

The feed is programmed under an F address except for rapid traverse. The unit may be in mm per minute or in mm per revolution. The unit of the federate has to be defined at the beginning of the programme. The feed rate can be calculated by the following formula in mm/min:

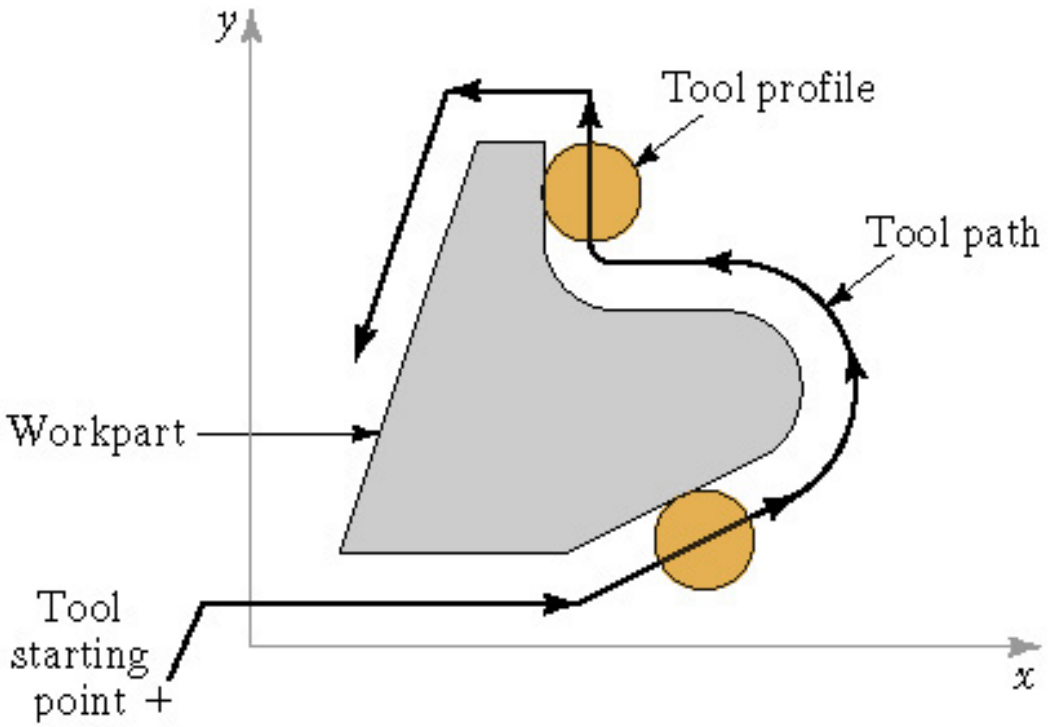
$$\text{Feet Rate} = \frac{\text{Chip Load}}{\text{Tooth} \times \text{No. of tooth} \times \text{Spindle speed}}$$

Tool Function

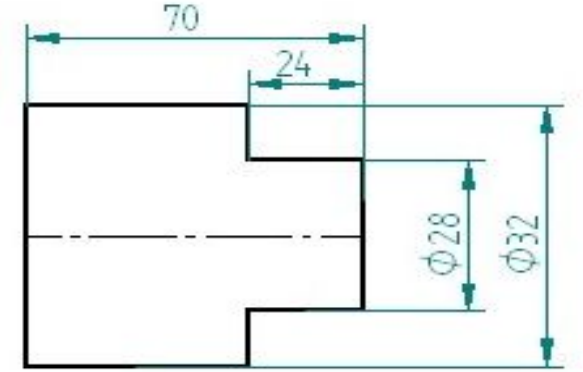
The selection of tool is commanded under a T address. T04 represents tool number 4.

Cutter Offset

Cutter path must be offset from actual part outline by a distance equal to the cutter radius

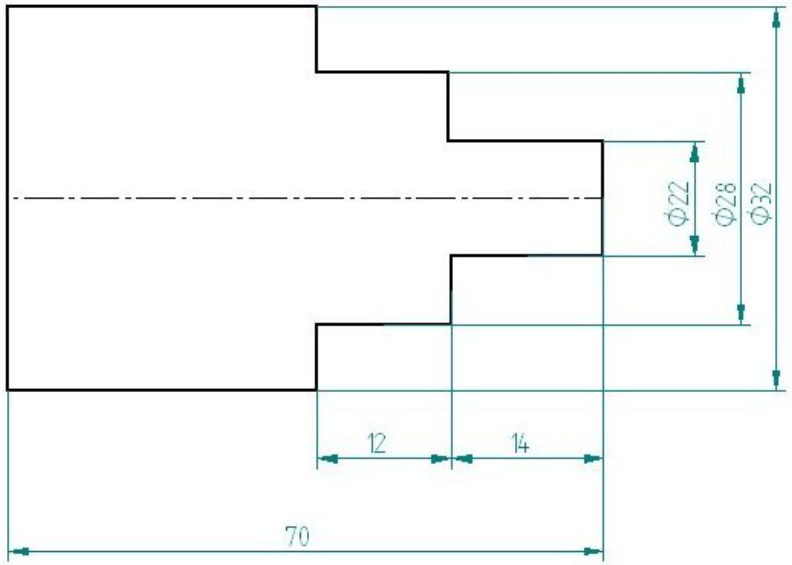


Generate program for turning operation for the profile given below. Cutting tool: RH turning tool.




```
G21 G98 G18
G28 U0 W0
M06 T01
M03 S1500
G00 X32 Z2
G01 X31 F100;
G01 Z-24;
G01 X32;
G01 Z0
G01 X30
G01 Z-24
G01 X32
G00 Z5
G01 X29
G01 Z-24
G01 X32
G00 Z5
G01 X28
G01 Z-24
G01 X32
G00 Z5
G28 X0 Z0
M05
M30
```

Generate program for turning operation for the profile given below. Cutting tool: RH turning tool.



Turning: G90 Cycle

Syntax

G90 X Z F

Where,

X – Required diameter or Depth of cut

Z - Turning Length

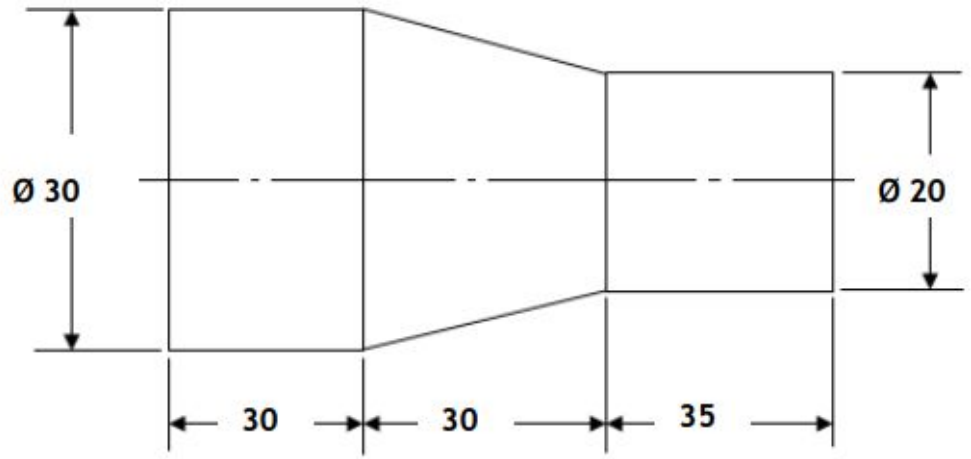
F - Feed rate

```
G21 G98 G18
G28 U0 W0
M06 T01
M03 S1500
M07
G00 X32 Z2
G01 Z0 F80
G90 X31 Z-26 F80
G01 X30
G01 X29
G01 X28
G01 X27 Z-14
G01 X26
G01 X25
G01 X24
G01 X23
G01 X22
G28 U0 W0
M05
M08
```

```
G21 G98
G28 U0 W0
M06 T01
M03 S1500
G00 X32 Z05
G90 X31 Z-26 F100
G01 X30
G01 X29
G01 X28
G01 X27 Z-14
G01 X26
G01 X25
G01 X24
G01 X23
G01 X22
G01 X32
G28 U0 W0
M05
M30
```

Example

02 (All dimensions are in mm).



G71 –TURNING CYCLE PROGRAM

G71 U R

G71 P Q U W F

U-DEPTH OF CUT AT EACH PASS

R- RETRACTION

P- STARTING BLOCK OF THE PROFILE

Q- END BLOCK OF THE PROFILE

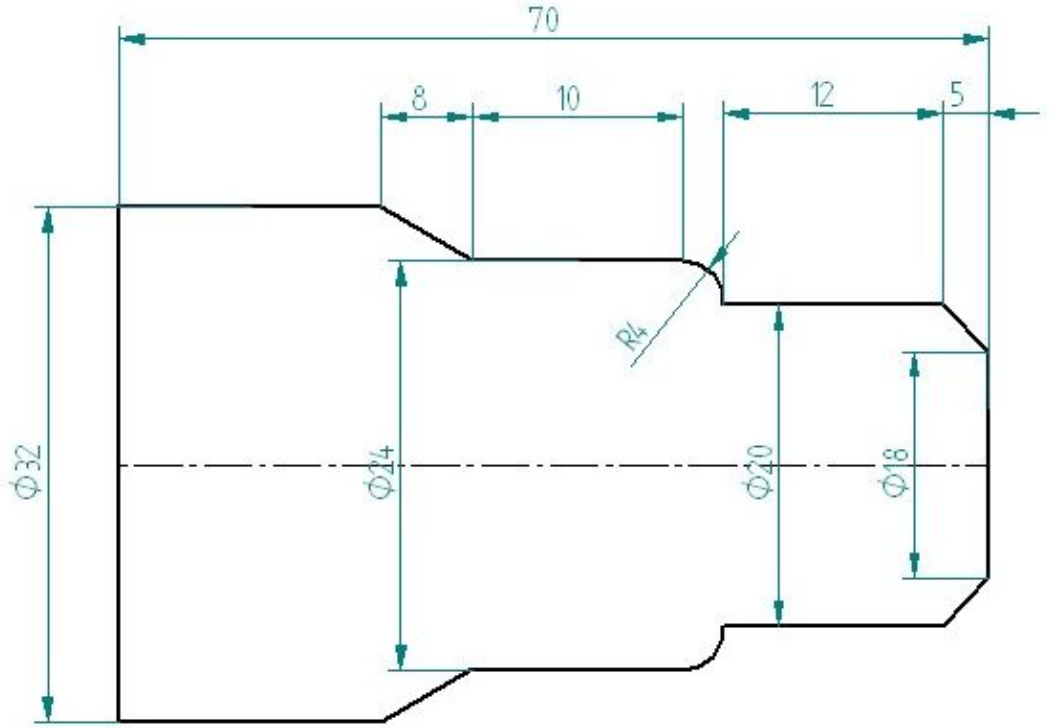
U- FINISHING ALLOWANCE IN X AXIS

W- FINISHING ALLOWANCE IN Z AXIS

F- FEED RATE

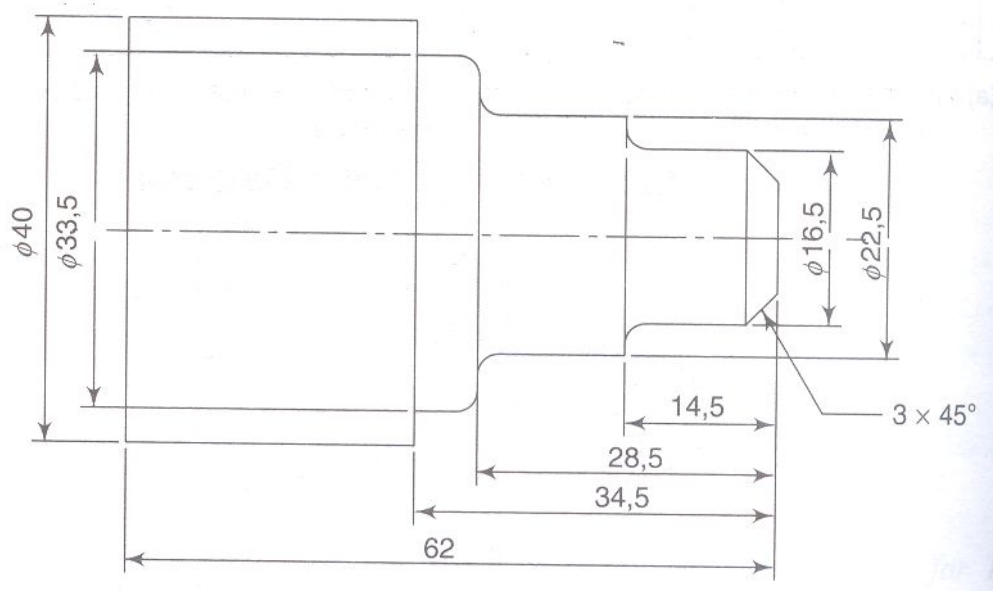
```
PROGRAM
G21 G98
G28 U0 W0
M06 T01
M03 S1000
G00 X30 Z05
G71 U1 R0.5
G71 P10 Q20 U0 W0 F80
N10 G01 X20 F80
      G01 Z-35
N20 G01 X30 Z-65
G01 X32
G28 U0 W0
M05
M30
```

Generate program for turning operation for the profile given below. Cutting tool: RH turning tool



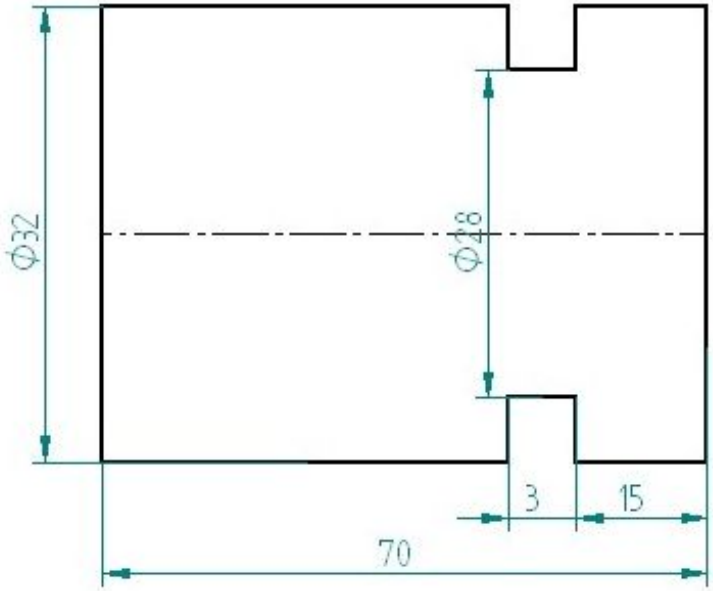

```
G21 G98
G28 U0 W0
M06 T02
M03 S1000
G00 X32 Z2
G01 Z0 F80
G71 U1 R0.5
G71 P10 Q20 U0.1 W0.1 F80
N10 G01 X18
G01 Z0
G01 X20 Z-5
G01 Z-17
G03 X24 Z-21 R4
G01 Z-31
N20 G01 X32 Z-39
G70 P10 Q20 U0 W0 F80
G28 U0 W0
M05
M30
```

Generate program for turning operation for the profile given below. Cutting tool: RH turning tool




```
G21 G98
G28 U0 W0
M06 T01
M03 S1000
G00 X40 Z5
G71 U1 R0.5
G71 P1 Q2 U0.1 W0.1 F100
N1 G01 X13.5 F100
G01 Z0
G01 X16.5 Z-3
G01 Z-14.5
G01 X22.5
G01 Z-28.5
G01 X33.5
G01 Z34.5
N2 G01 X40
G70 P1 Q2 S1500 F80
G28 U0 W0
M05
M30
```

Grooving cycle




```
G21 G98
G28 U0 W0
M06 T01
M03 S1000
G00 X33 Z-17
G01 X32.5 F100
G75 R1
G75 X28 Z-18 P100 Q1000 F40
G28 U0 W0
M05
M30
```

Grooving: G75 Cycle

Syntax

G75 R

G75 X Z P Q F

Where,

R - Retraction

X - Required Diameter

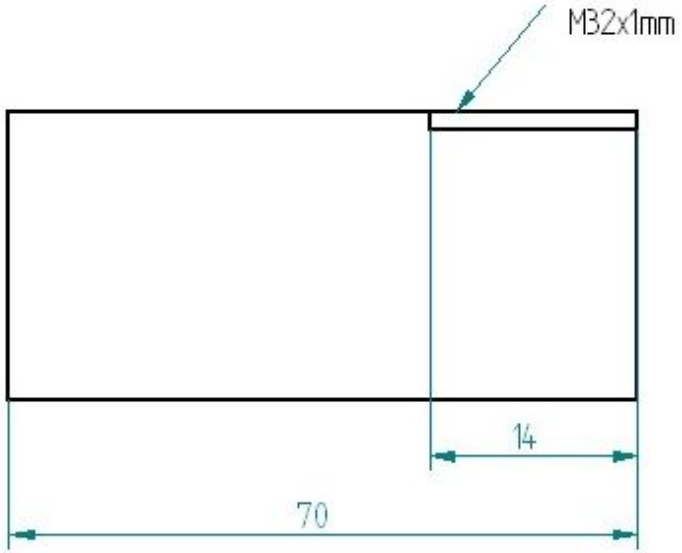
Z - Grooving Distance

P - Depth of Cut in X axis (in micron)

Q - Tool Movement in Z axis (in micron)

F - Feed rate

Generate program for threading operation for the profile given below. Cutting tool: RH turning tool.



Thread Cutting: G76 Cycle

Syntax

G76 P(_A_) (_B_) (_C_) Q R

G76 X Z P Q F

Where,

P(_A_) - No. of Finishing passes

P(_B_) - Thread Chamfer

P(_C_) - Thread Angle

Q - Depth of cut (in micron)

R - Finishing Allowance

X - Minor diameter

Z - Thread length

P - Thread height (in micron)

F - Pitch of the thread

To find minor diameter

d = D - 2P

P = 0.613 * pitch of the thread

D=30.774

Thread cutting

```
G21 G98
G28 U0 W0
M06 T01
M03 S500
G00 X33 Z5
G01 Z1 F100
G76 P051560 Q50 R0.01
G76 X30.774 Z-14 P0613 Q150 F1
G28 U0 W0
M05
M30
```

Generate program for turning operation for the profile given below. Cutting tool: RH turning tool

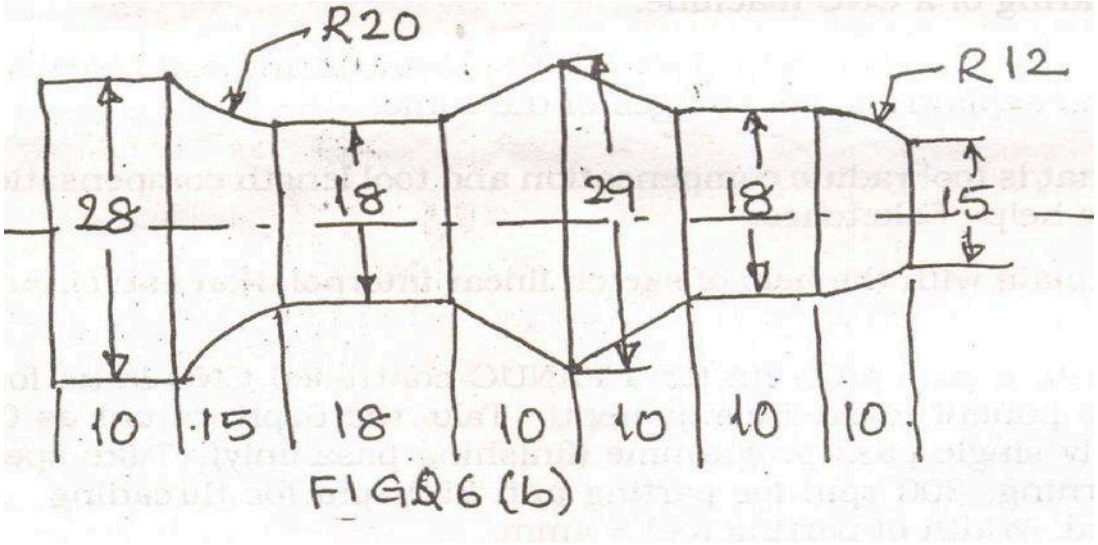
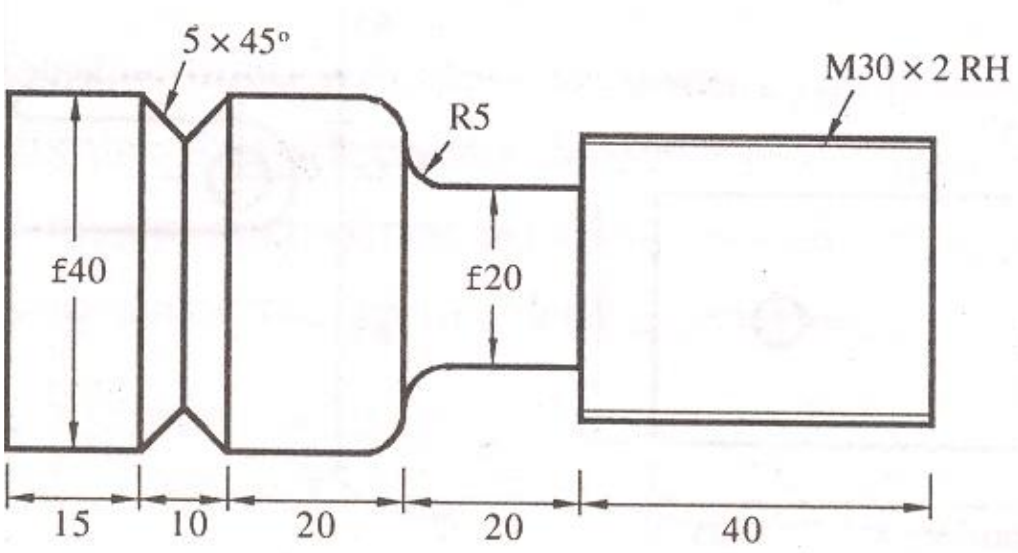


FIG Q 6 (b)

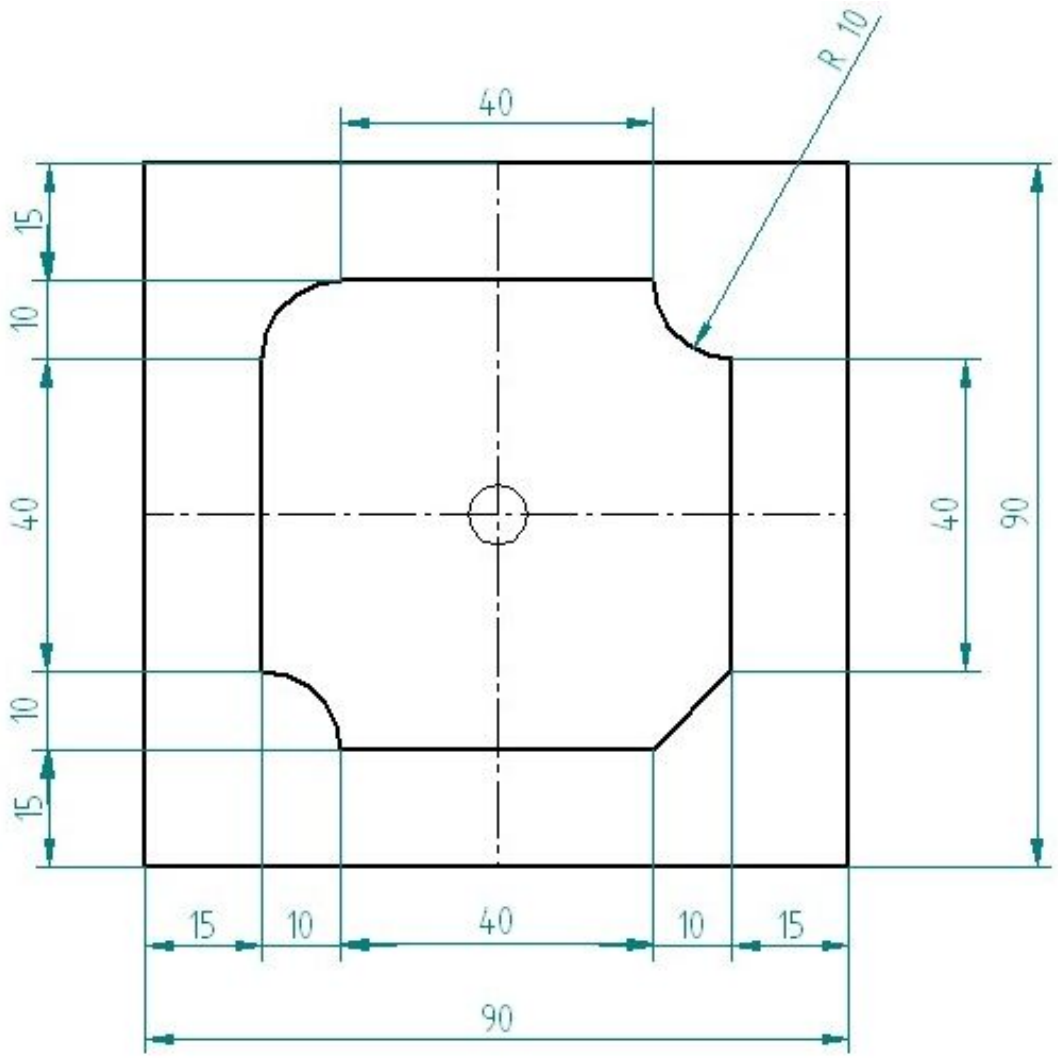
```
G21 G98
G28 U0 W0
M06 T01
M03 S1000
G00 X28 Z2
G01 Z0 F100
G71 U0.5 R1
G71 P10 Q20 U0 W0 F80
N10 G01 X15
G03 X18 Z-10 R12
G01 Z-20
G01 X29 Z-30
G01 X18 Z-40
G01 Z-58
N20 G01 X28 Z-73 R20
G28 U0 W0
M05
M30
```

Generate program for turning operation for the profile given below.
Cutting tool: RH turning tool




```
G21 G98
G28 U0 W0
M06 T01
M03 S1000
G00 X28 Z2
G01 Z0 F100
G71 U0.5 R1
G71 P10 Q20 U0 W0 F80
N10 G01 X30
G01 Z-40
G01 X20
G01 Z-55
G03 X_ Z_ R_
G01
G01
N20 G01
G28 U0 W0
M03
M01
M06 T02
M03 S500
G00 X30 Z0
G76 P041560 Q_ R0.01
G76 X_  Z-40 P_ Q_ F2
G28 U0 W0
M05
M30
```

Generate milling program and run the simulation for the profile given below. Cutting tool: Slot Drill Cutter (ϕ 6mm), Depth of cut: 1mm



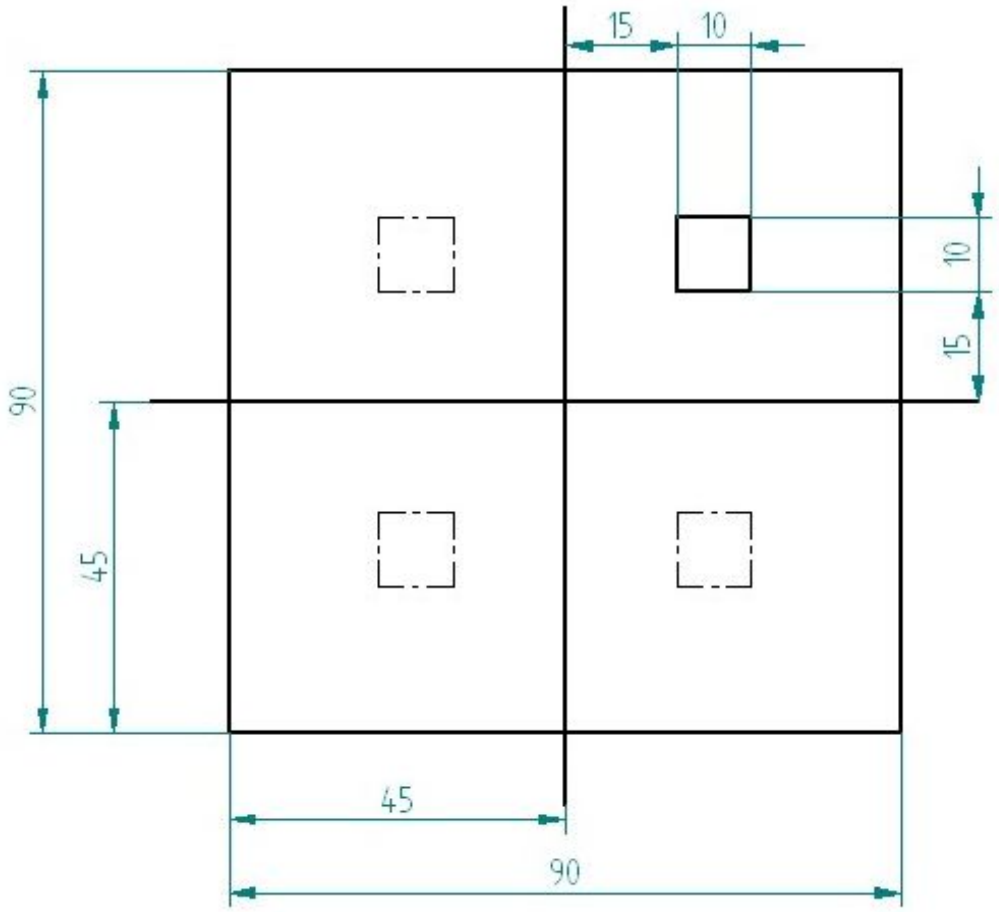
1	15	25
2	15	65
3	25	75
4	65	75
5	75	65
6	75	25
7	65	15
8	25	15
9	15	25

```
G17 G40 G94 G21
G28 U0 W0
M06 T01
M03 S1000
G00 X0 Y0 Z2
G41
G01 X15 Y25 F80
G01 Z-2
G01 X15 Y65
G02 X25 Y75 R10
G01 X65 Y75
G03 X75 Y65 R10
G01 X75 Y25
G01 X65 Y15
G01 X25 Y15
G03 X15 Y25 R10
G01 Z2
G28 U0 W0
M05
M30
```

1	15	25
2	15	65
3	25	75
4	65	75
5	75	65
6	75	25
7	65	15
8	25	15
9	15	25

Mirroring program

Generate milling program and run the simulation for the profile given below. Cutting tool: Slot Drill Cutter (ϕ 6mm), Depth of cut: 2mm



M70- X MIRROR ON
M71- Y MIRROR ON
M80- X MIRROR OFF
M81- Y MIRROR OFF

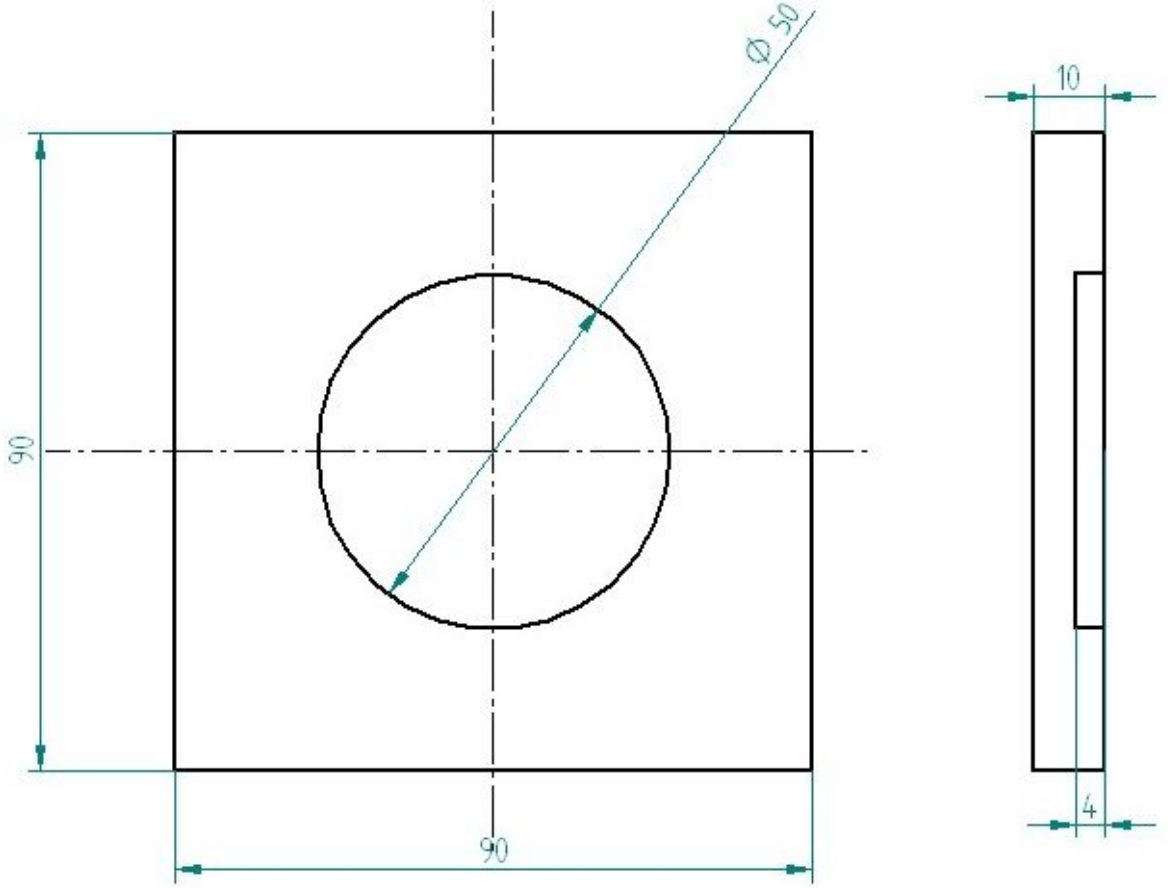
M98- CALLING FOR SUB PROGRAM
M99- END OF SUBPROGRAM

```
G17 G40 G94 G21
G28 U0 W0
M06 T01
M03 S1000
G00 X0 Y0 Z2
M98 P8888
M70
M98 P8888
M80
M71
M98 P8888
M81
M70
M71
M98 P8888
M80
M81
G00 Z2
G28 U0 W0
M05
M30
```

```
:8888
G01 X15 Y15 F100
G01 Z-2
G01 X15 Y25
G01 X25 Y25
G01 X25 Y15
G01 X15 Y15
G01 Z5
M99
```

Circular Pocket Milling

Generate milling program and run the simulation for the profile given below. Cutting tool: Slot Drill Cutter (ϕ 6mm).



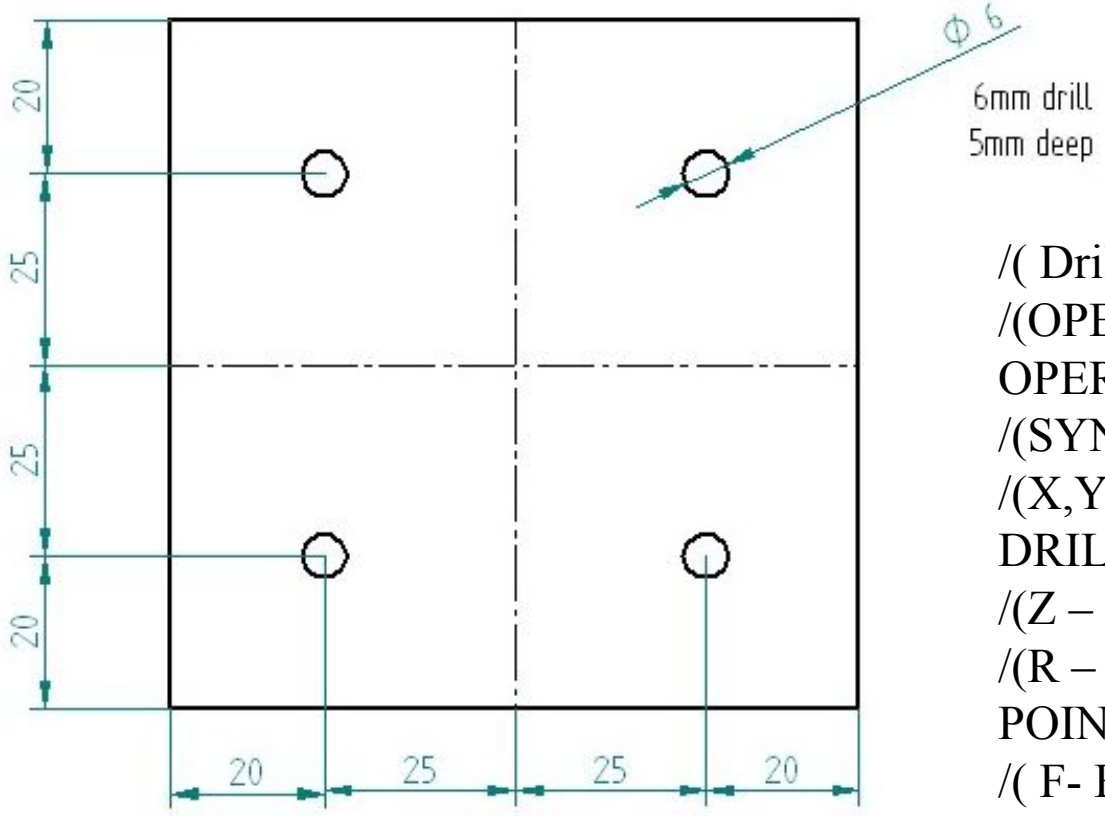
CIRCULAR POCKET OPERATION)
/(G170 P Q R X Y Z I J K)
/(G171 P S R F B J)
/(P0 – ROUGHING OPERATION)
/(P1 - FINISHING OPERATION)
/(Q – DEPTH OF CUT FOR EACH PASS)
/(R – REFERENCE POINT)
/(X – POSITION OF TOOL IN X-AXIS)
/(Y – POSITION OF TOOL IN Y-AXIS)
/(Z – TOTAL DEPTH)
/(I – SIDE FINISHING ALLOWANCE)
/(J – BASE FINISHING ALLOWANCE)
/(K – RADIUS OF THE POCKET)
/(P – CUTTER MOVEMENT IN%, I.E, 50% OF THE CUTTER DIAMETER)
/(S – ROUGHING OPERATION SPINDLE SPEED)
/(R – ROUGHING OPERATION FEED IN Z-AXIS)
/(F – ROUGHING OPERATION FEED IN X&Y AXIS)
/(B – FINISHING OPERATION SPINDLE SPEED)
/(J – FINISHING OPERATION FEED IN X,Y &Z AXES)

```
G17 G40 G94 G21
G28 U0 W0
M06 T01
M03 S2000
G00 X0 Y0 Z2
G170 P0 Q1 R0.5 X0 Y0 Z-4 I0.1 J0.1 K25
G171 P50 S1500 R30 F80 B2000 J30
G170 P1 Q1 R0.5 X0 Y0 Z-4 I0 J0 K25
G171 P50 S1500 R30 F80 B2000 J30
G00 Z5
G28 U0 W0
M05
M30
```

Generate a CNC program for performing drilling operation and simulate for the profile given below.

Hole diameter 6mm and depth 5mm.

Cutting tool : Slot drill cutter of 6mm diameter.



/(Drilling operation: Dia.6mm drill)
/(OPERATION #1: DRILLING OPERATION)
/(SYNTAX:G73 X Y Z Q R F K)
/(X,Y – COORDINATE POINT OF DRILLING
/(Z – DEPTH OF DRILL HOLE)
/(R – RETURN TO REFERENCE POINT)
/(F- FEED RATE)
/(K-NUMBER OF DRILL HOLES)
Q- DEPTH OF CUT AT EACH PASS)

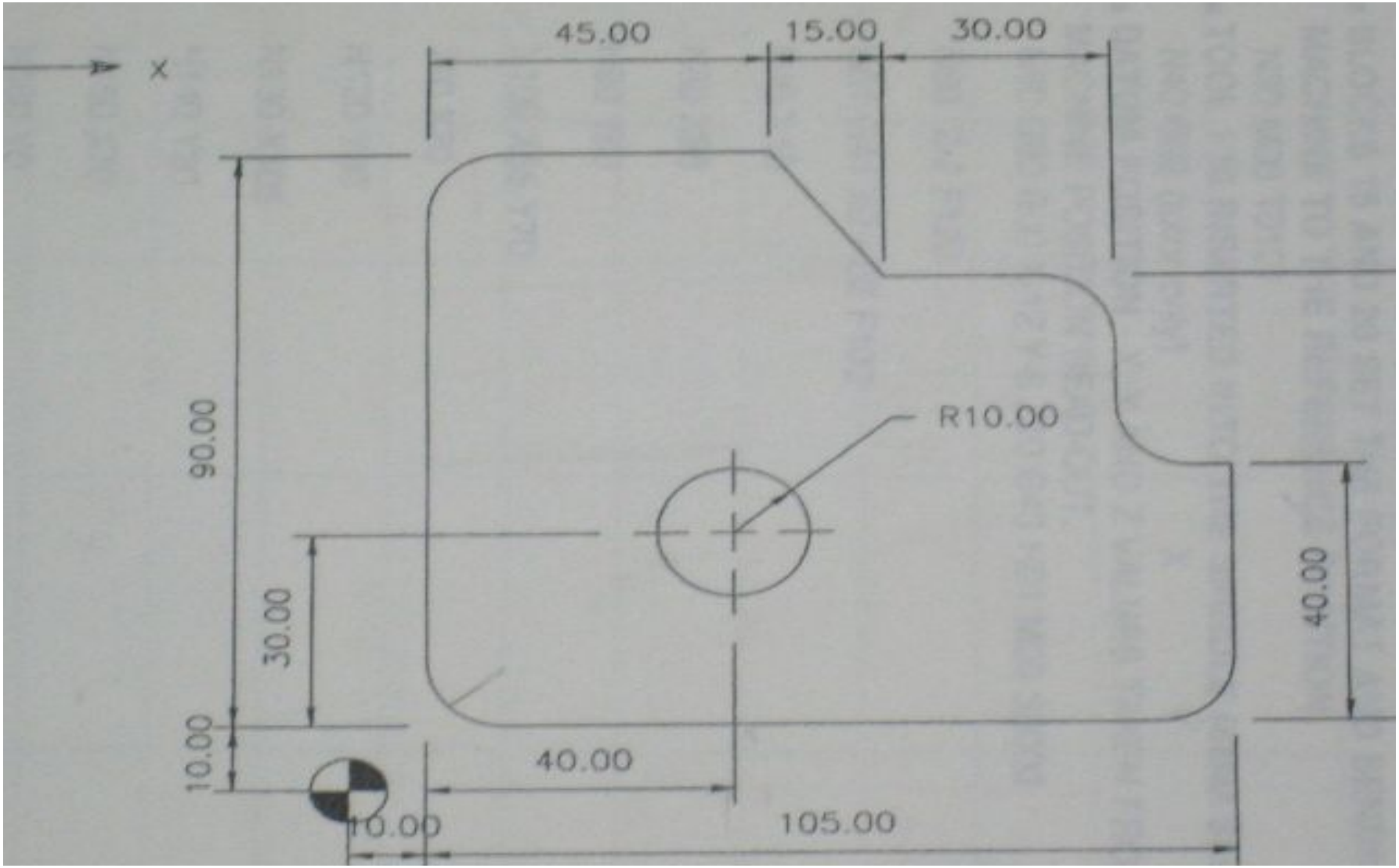
G73 DRILLING CYCLE

G73 X Y Z Q R F K

```
G17 G40 G94 G21 G90
G28 U0 W0
M06 T01
M03 S1000
G00 X0 Y0 Z2
G73 X25 Y25 Z-5 Q1 R0.5 F50 K4
G01 X25 Y-25
G01 X-25 Y-25
G01 X-25 Y25
G01 Z2
G28 U0 W0
M05
M30
```

Generate a CNC program for performing drilling operation and simulate for the profile given below.

Cutting tool : Slot drill cutter of 6mm diameter.



/(Drilling operation: Dia.6mm drill)
/(OPERATION #1: DRILLING OPERATION)
/(SYNTAX:G73 X Y Z Q R F K)
/(X,Y – COORDINATE POINT OF DRILLING
/(Z – DEPTH OF DRILL HOLE)
/(R – RETURN TO REFERENCE POINT)
/(F- FEED RATE)
/(K-NUMBER OF DRILL HOLES)

Automatically Programmed Tool (APT)

The APT language was the product of the MIT developmental work on NC programming systems. Its development began in June 1956, and it was first used in production around 1959. Today it is the most widely used language in the United States. Although first intended as a contouring language, modern versions of APT can be used for both positioning and continuouspath programming and continuous-path programming in up to five axes

Automatically Programmed Tool (APT)

APT is not only a language; it is also the computer program that processes the APT statements to calculate the corresponding cutter positions and generate the machine tool control commands. To program in APT. the part geometry must first be defined. Then the tool is directed to various point locations and along surfaces of the workpart to accomplish the required machining operations. The viewpoint of the programmer is that the workpiece remains stationary and the tool is instructed to move relative to the part. To complete the program, speeds and feeds must be specified, tools must be called, tolerances must be given for circular interpolation, and so forth.

Four basic types of statements in the APT language

1. Geometry statements, also called definition statements, are used to define the geometry elements that comprise the part.
2. Motion commands are used to specify the tool path.
3. Post-processor statements control the machine tool operation, for example, to specify speeds and feeds, set tolerance values for circular interpolation, and actuate other capabilities of the machine tool.
4. Auxiliary statements, a group of miscellaneous statements used to name the part program, insert comments in the program and accomplish similar functions.

Geometry Statements. The geometry of the part must be defined to identify the surfaces and features that are to be machined. Accordingly, the points, lines, and surfaces must be defined in the program prior to specifying the motion statements. The general form of an APT geometry statement is the following:

Symbol= Geometry type/ descriptive data
P1 = POINT/5.0,4.0,0.0

To specify a point
P2 = PONIT/INT OF, L1, L2
To specify A line
L3 = LINE/ P3, P4
L4 = LINE/ P5, PARLEL, L3

Points.

P1 = POINT/5.0,4.0,0.0

where the descriptive data following the slash indicate the *X·, yo,* and z coordinates. The specification can be done in either inches or millimeters (metric). We use metric values in our examples. As an alternative, a point can be defined as the intersection of two lines, as in the following

P2 = POINT/INTOF, Lt. L2

where the APT word INTOF in the descriptive data stands for "intersection of." Other methods of defining points are given in the Appendix under POINT

Lines. A line defined in APT is considered to be of infinite lengtb in hoth directions. Also, APT treats a line as a vertical plane that is perpendicular to the .rj' plane. The easiCStway to specify a line is by two points through which it passes:

L3 = LINE/P3. P4

In some situations, the part programmer may find it more convenient to define a new line as being parallel to another line that has been previously defined; for example,

L4 = LlNE/P5, PARLEL, L3

where PARLEL is APT's way of spelling "parallel.“ The statement indicate, line l4 passes through point PS and is parallel to line L3.

Planes. A plaue can be defined by specifying three points through which the plane passevasin the following'

PL1 = PLANE/Pl,P2,P3

Of course. the three points must he noncollinear. A plane can also be defined as being parallel to another plane that has been previously defined; for instance,

PL2 = PLANE/P2, PARLEL, PL1

which states that plane PL2 passes through point P2 and is parallel to plane PLI. In APT, a plane extends indefinitely.

Circles. In APT, a circle is considered to be a cylindrical surface that is perpendicular to the xj plane and extends to infinity in the z direction. The easiest way to define a circle is by its center and radius. as in the following

Cl = CIRCLE/CENTER,Pl,RADIUS,25.0

By convention. the circle is located in the .e.y plane. An alternative way of defining a circle i~to specify thur it passes through three points; for example,

C2 = CIRCLE/P4, P5, P6

Rules for formulating APT geometry statement

- Co-ordinate data must be specified in the order x,y,z
- Any symbol used as descriptive data must have been previously defined
- A symbol can be used to define only one geometry element
- Lines defined in APT are of infinite length in both directions similarly, planes extend indefinitely and circles defined in APT are complete circle.

Motion Commands

Motion Commands. All APT motion statements follow a common format, just as geometry statements have their own format. The format of an APT motion command is'

MOTION COMMAND/descriptive data

An example of an APT motion statement is

GOTO/PI

The statement consists of two sections separated by a slash. The first section is the basic command that indicates what move the tool should make. The descriptive data following the slash tell the tool where to go. **In** the above example, the tool is directed to go to (GOTO) point PI, which has been defined in a previous geometry statement.

Motion Commands

At the beginning of the sequence of motion statements, the tool must be given ll starting point. This is likely to be the target point. the location where the operator has positioned the tool at the start of the job. The part programmer keys into this starting position with the following statement:

FROM/PTARG
FROM/-2.0, -2.0, 0.0

PTARG-Target point or the location where the operator has positioned the tool at the start of the job.

Motion Commands

1.POINT TO POINT Statements

In our discussion of APT motion statements, it is appropriate to distinguish between point to point motions and contouring motions. For point to point *motions,* there are only two commands: GOTO and GODLTA. The GOTO statement instructs the tool to go to a particular point location specified in the descriptive data. Two examples are:

GOTO/P2

The GODLTA command specifies an incremental move for the tool. To illustrate, the following statement instructs the tool to move from its present position by a distance of 50.0 mm in the x direction, 120.0 mm in the y direction, and 40 mm in the z direction

GODLTA/50.0, 120.0, 40.0

The GODLTA statement is useful in drilling and related machining operations. The tool can be directed to go to a given hole location; then the GODLTA command can be used to drill the hole, as in the following sequence:

GOTO/P2

GODLTA/0, 0, 50.0

Motion Commands

Contouring motion commands:

Contouring motion commands are more complicated than PTP commands are because the tool's position must be continuously controlled throughout the move. To exercise this control. the tool is directed along two intersecting surfaces until it reaches a third surface. These three surfaces have specific names in APT; they are:

- Drive surface. This is the surface that guides the side of the cutter. It is pictured as II plane in our figure.
- Part surface. This is the surface, again pictured as a plane, on which the bottom or nose of the tool is guided.
- Check surface. This is the surface that stops the forward motion of the tool in the execution of the current command. One might say that this surface "checks" the advance of the tool.

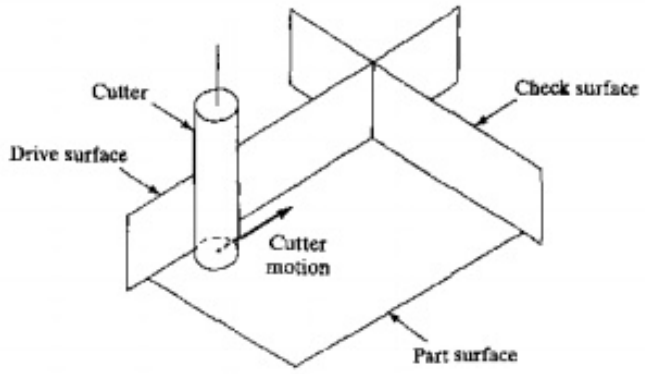
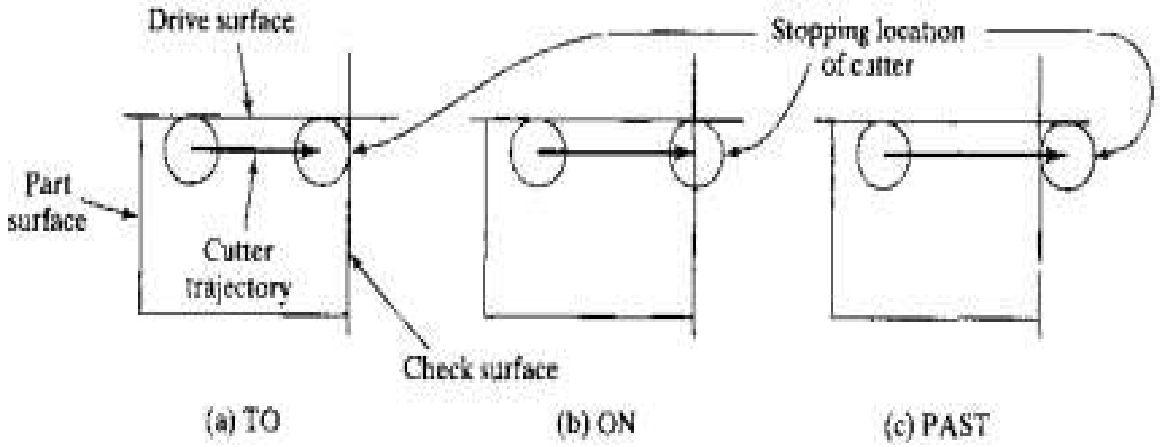


Figure 6.20 Three surfaces in APT contouring motions that guide the cutting tool.

The four modifier words arc TO, ON, PAST. and TANTO. As depicted in Figure, the word TO positions the leading edge of the tool in contact with the check surface; ON positions the center of the tool on the check surface; and PAST puts the tool beyond the check surface. so that its trailing edge is in contact with the check surface. The fourth modifier word TANTO is used when the drive surface is tangent to a circular check surface, TANTO moves the cutting tool to the point of tangency with the circular surface

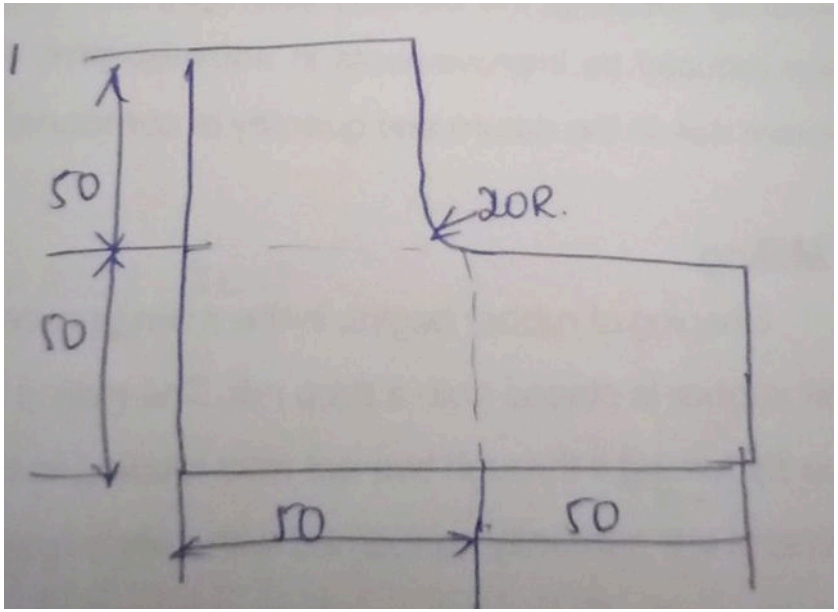


The six APT motion words
GOLFT, GORGT, GOFWD, GOBACK, GOUP, or GODOWN.
To begin with the sequence of motion commands the FROM statement is used
TARG statement target point where the operator has set up the tool

Post processor statements
COOLNT, MACHIN, TURRET, END, RAPID, FEDRAT, SPINDL

Auxiliary statements
CPRINT, CUTTER, FINI, INTOL, OUTTOL, PARTNO

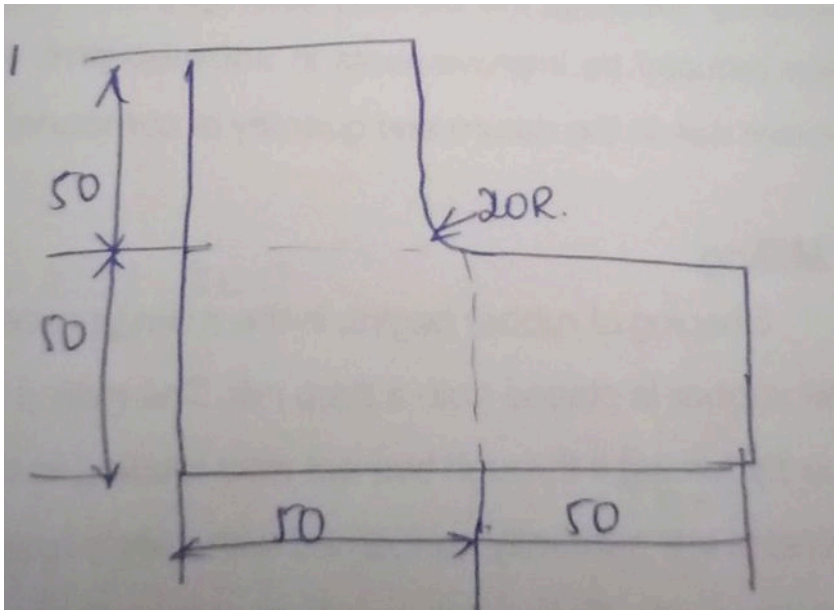
Write an APT program to machine the profile on the component shown, given is the following information

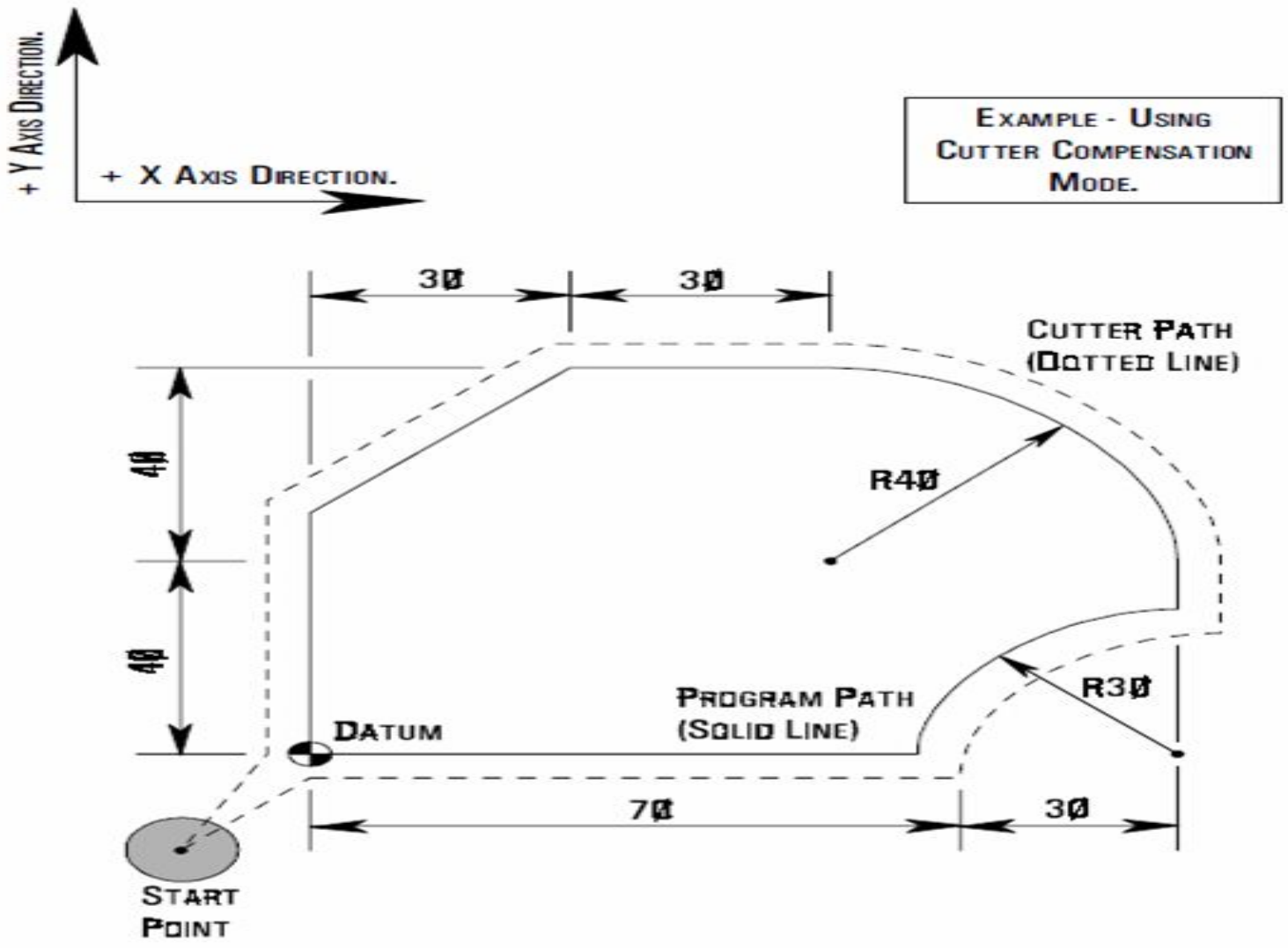


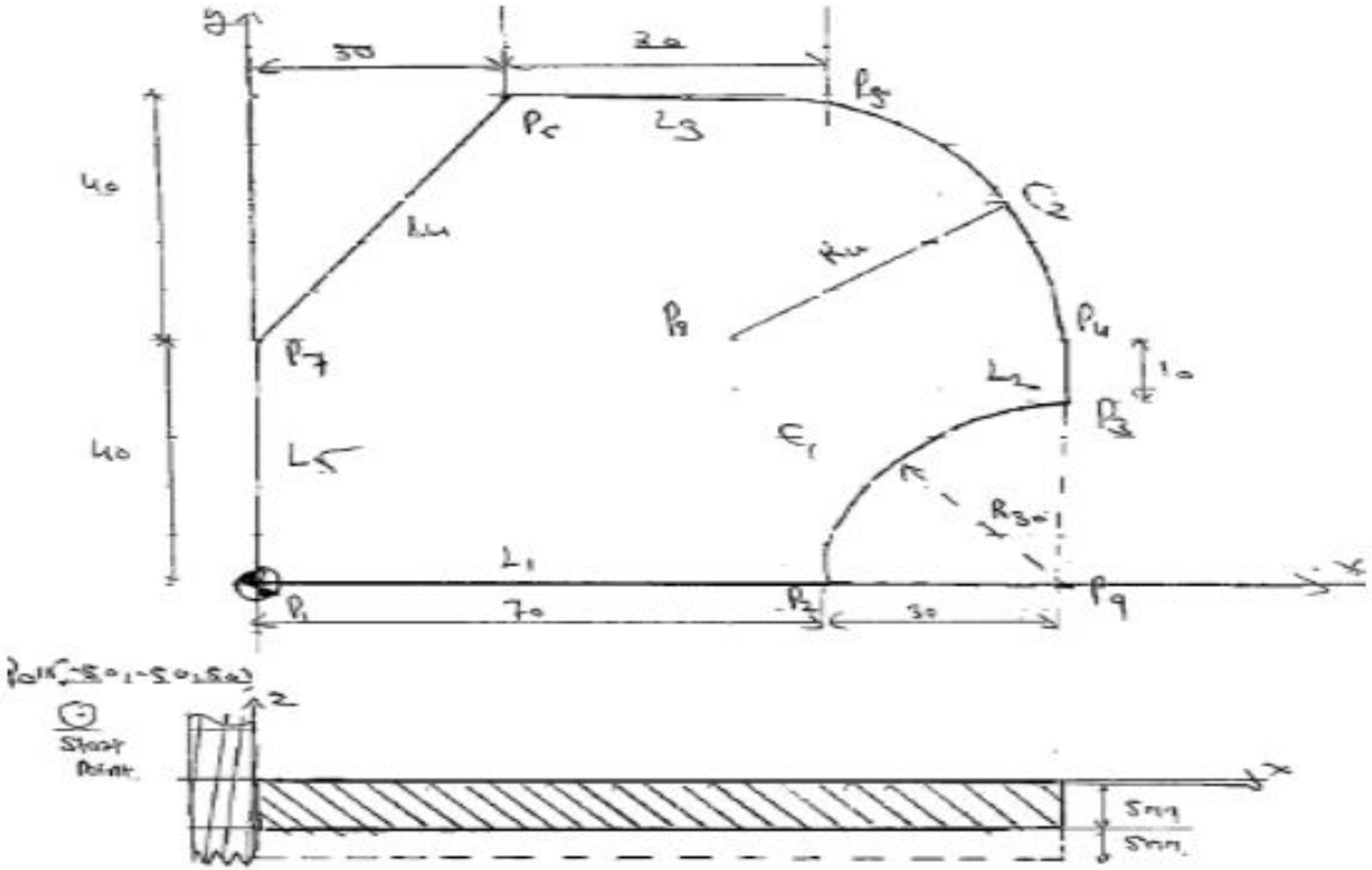
Post processor-MACHIN/MILL,1
Spindle speed – 1000rpm
Feed rate – 75mm/min
Tolerance – 0.01mm/min
Thickness- 15mm
Cutter dia – 20mm
Cutter location - -100.0, -100.0, 50.0

```
PART NO/1
MACHIN/MILL,01
INTOL/0.01
OUTOL/0.01
CUTTER/20.0
SPINDL/1000
FEDRAT/75
COOLNT/ON
STRT=POINT/-100.0, -100.0, 50.0
P0=POINT/0.0,0.0, 0.0
P1=POINT/100.0,0.0,0.0
P2=POINT/100.0,50.0,0.0
P3= POINT/70.0,70.0,0.0
P4= POINT/50.0,100.0,0.0
P5=POINT/0.0,100.0,0.0
L1=LINE/P0,P1
L2=LINE/P1, P2
L3=LINE/P2, PARLEL,L1
L4=LINE/P4, PERPTO, L1
L5=;INE/P4, PERPTO, L1
L6=LINE/P5, P0
C1=CIRCLE/CENTER, P3, RADIUS, 20.0
PL1=PLANE/P0,P1,P5
```

```
INTOL/0.01
OUTOL/0.01
CUTTER/20.0
SPINDL/1000
FEDRAT/75
COOLANT/ON
FROM / STRT
GO/TO, L1 TO PL1,TO L6
GORGT/L1, PAST, L2
GOLFT/L2, PAST, L3
GOLFT/L3, TANTO, C1
GOFWD/L4, PAST, L5
GOFWD/L5, PAST, L6
GOLFT/L6, PAST, L1
GOTO/P0
COOLANT/OFF
SPINDL/OFF
FINI
```




+ Y Axis Direction.
+ X Axis Direction.
Example - Using Cutter Compensation Mode.
30
30
Cutter Path (Dotted Line)
40
40
R40
R30
Program Path (Solid Line)
Datum
70
30
Start Point


30
2a
40
40
70
30
10
Shear Point
5mm
5mm

```
PART NO/1
MACHIN/MILL,01
INTOL/0.01
OUTOL/0.01
CUTTER/10.0
SPINDL/1000
FEDRAT/75
COOLNT/ON
P0=POINT/-50,-50,0
P1=POINT/0,0,-10
P2=POINT/70,0,-10
P3=POINT/100,0,-10
P4=POINT/100,40,-10
P5=POINT/60,80,-10
P6=POINT/30,80,-10
P7=POINT/0,40,-10
P8=POINT/60,40,-10
P9=POINT/100,0,-10
```

```
L1=LINE/P1, P2
L2=LINE/P3,P4
L3=LINE/P5, P6
L4=LINE/P6, P7
L5=LINE/P7, P1
C1=CIRCLE/CENTER, P9, RADIUS, 30
C2=CIRCLE/CENTER, P8, RADIUS, 40
PL1=PLANE/P1, P2, P3
```

```
SPINDL/1200
FEDRAT/960
COOLANT/ON
FROM/P0
GO/TO, L1 TO PL1, TO P5
GORGT/L1,PAST, P2
GOFWD/C1, PAST, L2
GOLFT/L2, PAST, P4
GOFWD/C2, TANTO, L3
GOLFT/L3, PAST, L4
GOLFT/L4,PAST, L5
GOLFT/L5, PAST, L1
GOTO/P0
SPINDL/OFF
COOLNT/OFF
FINI
```

PART PROGRAMMING USING CAD/CAM

A CAD/CAM system is a computer interactive graphics system equipped with software to accomplish certain tasks in design and manufacturing functions. One of the important tasks performed on a CAD/CAM system is NC part programming. In this method of part programming, portions of the procedure usually done by the part programmer are instead done by the computer. The two main tasks of a part programmer in a computer assisted programming are

a) defining the part geometry
b) specifying the tool path.

The proposed methodology is used to automate both of these tasks

PART PROGRAMMING USING CAD/CAM

Part geometry definition: The fundamental objective of CAD/CAM system is to integrate the design engineering and manufacturing engineering functions.

- If a CAD/CAM system is used, a computer graphics model of each part is developed by the designer and stored in the CAD/CAM database.
- That model contains all of the geometric, dimensional and material specifications for the part.
- When the same CAD/CAM system, or a CAM system that has access to the same CAD database in which the part model resides, is used to perform NC part programming it makes little sense to re-create the geometry of the part during the programming procedure.
- Instead, the programmer has the capability to retrieve the part geometry model from the storage and to use that model to construct the appropriate cutter path.

- The significant advantage of using CAD/CAM in this way is that it eliminates one of the time consuming steps in computer assisted part programming geometry definition.
- After the part geometry has been retrieved, the usual procedure is to label the geometric elements that will be used during part programming.
- These labels are the variable names (symbols) given to the lines circles and surfaces that comprise the part.

Tool path generation using CAD/CAM

The second task of the NC programmer in computer assisted part programming is tool path Specification.
The first step in specifying the tool path is to select the cutting tool for the operation. Most CAD/CAM systems have tool libraries that can be called by the programmer to identify what tools are available in the tool crib.
The programmer must decide which of the available tools is most appropriate for the operation under consideration and specify it for the tool path.
This permits the tool diameter and other dimensions to be entered automatically for tool offset calculations.
If the desired cutting tool is not available in the library, an appropriate tool can be specified by the programmer.
It then becomes part of the library for future use.

Tool path generation using CAD/CAM

- The next step is tool path definition. There are differences in capabilities of the various CAD/CAM systems, which result in different approaches for generating the tool path.
- The most basic approach involves the use of the interactive graphics system to enter the motion
- Commands one by one, similar to computer assisted part programming. Individual statements in APT or other part programming language are entered and the CAD/CAM system provides an immediate graphic display of the action resulting from the command, thereby validating the statement. A more advanced approach for generating tool path commands is to use one of the automatic software modules available on the CAD/CAM system.